
System and tool design for lunar regolith manipulation

Project report
ROB9 - Group 962



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STUDENT REPORT

Title:

System and tool design for lunar regolith manipulation

Theme:

Technological Project Work

Project Period:

September 2024 - December 2024

Project Group:

Group 962

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Copies: 1**Page Numbers:** 91**Date of Completion:**

20. September 2024

Abstract:

This report presents the design and development of a robotic system aimed at lunar surface preparation, focusing on regolith manipulation for landing site and habitat construction. The project leverages the RobuROC 4 platform integrated with SLAM (Simultaneous Localization and Mapping) algorithms and tooling to address challenges in autonomous navigation, regolith handling, and terrain adaptation in the Moon's unique environment. While the SLAM implementation was attempted, it did not achieve the intended functionality, and no navigation solution was implemented. Additionally, the system was not tested under simulated lunar conditions. A blade-based tool was conceptualized for excavating and ripping regolith, with implementation and evaluation conducted within constrained terrestrial settings. This work highlights the challenges and partial advancements in developing versatile and efficient robotic solutions for sustainable lunar habitation.

Aalborg University, 24. May 2022

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1 Introduction

Space exploration represents one of humanity's most significant achievements, driven by the desire to extend our reach beyond Earth. Since the final lunar landing in 1972, when Eugene Cernan became the last person to walk on the Moon, attention shifted from manned missions to scientific objectives, including robotic missions, space telescopes, and satellite deployment to study the solar system and cosmos [1]. However, recent years have renewed interest in manned space exploration, with plans to establish a sustained human presence in space, as the one illustrated on Figure 1.1. As NASA and other space agencies aim to send astronauts to Mars, the Moon has become a focal point for developing technologies and extracting resources. Launching missions from the Moon requires less energy due to its lower gravity, and lunar resources can be used for fuel production, making it a strategic platform for deeper space exploration. [2]–[4]



Figure 1.1: Illustration of Lunar habitat, taken from [5]

Turning this vision into reality, as outlined in ESA [2], [6] and NASA [3], [4] roadmaps, requires overcoming technical challenges. One critical task is lunar surface preparation for landing sites and habitats. The Moon's regolith, a fine, abrasive dust, poses risks to spacecraft, structures, and equipment. Safe landings and stable foundations for lunar infrastructure demand surface modification techniques. Advanced robotic systems will be essential for preparing the surface, laying infrastructure, and maintaining facilities in the hazardous lunar environment.

This report focuses on the challenges of lunar surface preparation, a crucial area that intersects with multiple mission objectives and can be adapted for other scientific purposes when not in use for construction.

1.1 Initial problems statement

The need for robotic solutions in lunar exploration and development is evident. To support future missions, these systems must function autonomously, reducing human exposure to the hazards of the lunar environment. Moreover, with the Moon designated as a site for developing technologies and research that will advance future space exploration, these robotic systems must be capable of operating effectively in its challenging conditions.

”How can an autonomous robotic system be developed for deployment on the lunar surface to prepare the lunar regolith for construction and landing sites, enabling sustainable exploration and future human missions?”

2 Problem analysis

Lunar space exploration is entering a new era, with international efforts aimed at establishing a permanent presence on our nearest celestial neighbour; the Moon. NASA's Artemis program plays a central role in this endeavour, with its ambitions of returning humans to the Moon and laying the groundwork for more permanent facilities. Artemis aligns with the broader vision set out in NASA's Moon to Mars exploration roadmap, which outlines plans for the development of technologies essential to long-term lunar habitation. Similarly, ESA has detailed its own lunar exploration roadmap, which includes contributions to sustainable infrastructure, resource utilisation and international partnerships. [4], [7], [8]

This analysis explores the current need for extraterrestrial technologies, the challenges they face in development, deployment, and operation, and examines the existing technologies that are paving the way for future space exploration. The following sections outline the required task scope from both NASA and ESA. Next, the challenges associated with these tasks are investigated, followed by a review of related works addressing one or some of these challenges.

2.1 Task scope

The development of lunar infrastructure is central topic in both NASA and ESA space exploration roadmaps. Their roadmaps outline long-term plans for building sustainable habitats and research facilities, enhancing future robotic and human missions. Specifically, NASA's Moon to Mars architecture anchored by the Artemis missions, aims to establish a long-term human presence on the lunar surface. Furthermore, it aims at using the Moon as a steppingstone for deeper space exploration. Core infrastructure priorities include surface power generation, in-situ resource utilisation (ISRU) and construction. A growing focus is placed on autonomous robotic systems for preparing the lunar surface for landing zones, extracting and using materials, and future structures. [2]–[4], [7], [8]

ESA's Terrae Novae [2] architecture and Moonlight [9] program aim to establish a lasting European presence in space and on the lunar surface, while enabling communication and navigation capabilities. Terrae Novae seeks to strengthen Europe's role in space, while Moonlight focuses on deploying satellites around the Moon to provide centralized communication and GPS-like navigation. ESA and NASA are collaborating closely to achieve these objectives. Both agencies benefit from development of autonomous surface preparation solutions, which will enable rapid and efficient development of lunar infrastructure. ESA's Moonlight program further supports this with its robust communication and navigation networks, enhancing both precision and coordination of robotic systems. Collectively, these initiatives lay the groundwork for elevating technological readiness levels (TRLs), critical for future deep-space missions. [2], [9], [10]

NASA, ESA and other space agencies have provided preliminary requirements on various aspects of lunar exploration and habitation, with a strong focus on establishing foundational infrastructure on the Moon. The following Figure 2.1 illustrates key tasks involved in site preparation for lunar bases, landing sites, etc.

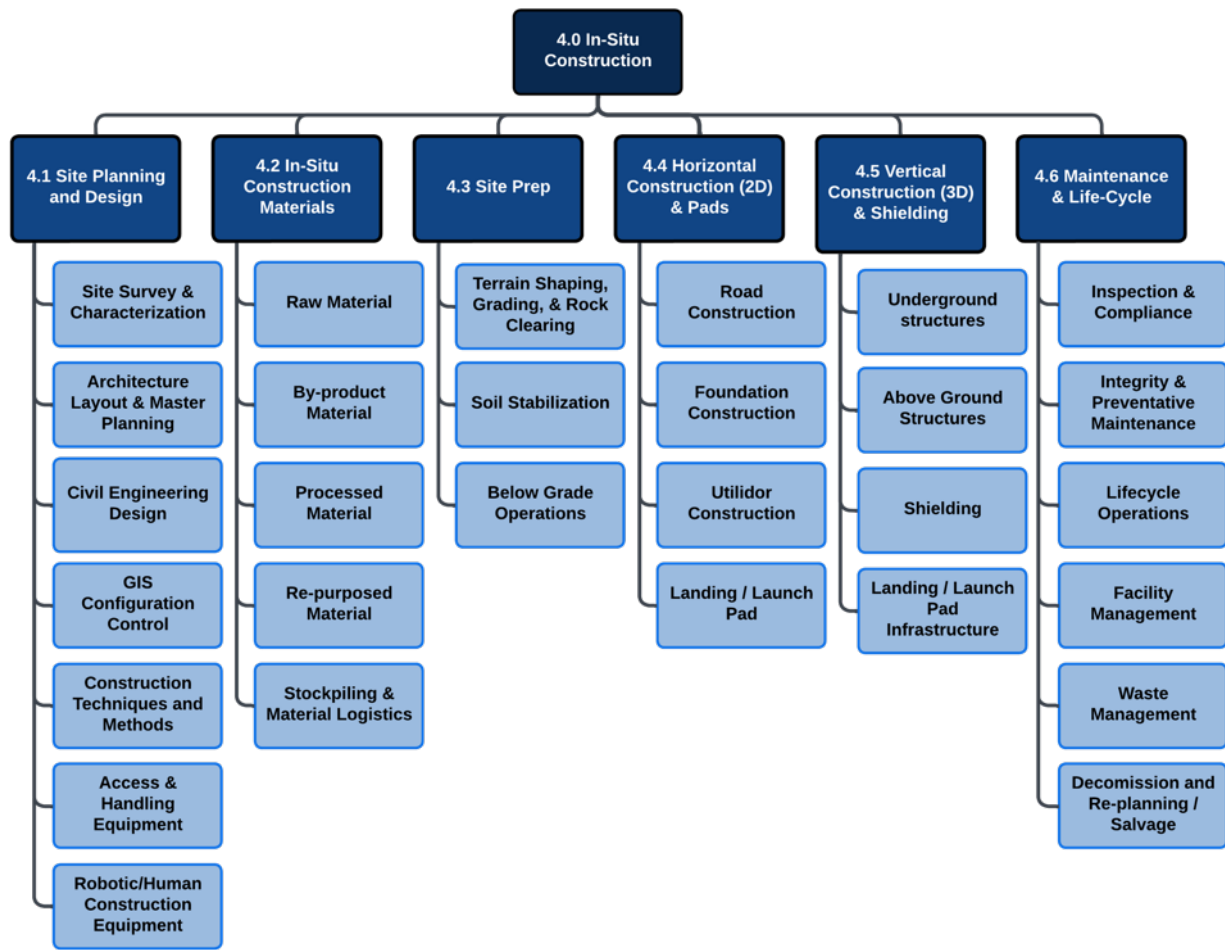


Figure 2.1: Illustration of work breakdown structure for infrastructure construction on the Moon, adapted from [11]

These tasks along with the consideration specified by Jehn et al. [12] defines the area in which improvements and development is needed. This project specifically focuses on the site preparation phase, which involves tasks such as:

- **Surface Area Coverage:** The robotic system must autonomously prepare at least $10m^2$ of lunar surface within a single operational cycle, ensuring it meets the demands of large-scale construction after several cycles[13].
- **Surface Compaction:** The robotic system must be able to compress the lunar regolith to a 90% compaction at 30cm depth, in order to create a stable surface for buildings and landing zones. [13]
- **Leveling Precision:** The system must achieve a surface flatness tolerance of less than 2 cm over $10m^2$ to meet NASA's landing pad construction requirements. [13]
- **Rock Detection and Removal:** The system should autonomously detect and relocate or collect rocks larger than 15 cm in diameter within its operational range. Collected rocks should be allocated for ISRU. [13]

These requirements along with the lunar exploration roadmaps, form the backbone for developing the an advanced robotic system for sustainable lunar habitation and exploration. Furthermore, based

on these tasks and requirements, the tasks that the system must be able to perform are categorised as: Exploration, Excavation, Compaction, Dust mitigation and Clean-up. Exploration covers the systems ability to explore the area in which work is to be performed. Excavation covers the ability to move and/or transport regolith and compaction is the systems ability to flatten and compact the regolith. Dust mitigation is the ability of the system to reduce dust plumes and regolith interference with operations. Lastly clean-up covers the ability to remove and/or transport rocks and debris away from the area of operation. However, these tasks still have challenges that need to be addressed.

2.2 Lunar Environment Challenges

The lunar environment poses significant challenges for construction and exploration, making it a vastly different experience compared to habitat building and landing surface preparation on Earth. This section discusses the abrasive nature of the lunar regolith, the complexities of the terrain, and the stability of the soil, all of which impact robotic operations on the lunar surface. Furthermore, it explores surface preparation standards essential for establishing lunar habitats and landings pads, as well as the restricted usage of certain sensor technologies, due to the Moon's unique conditions.

2.2.1 Terrain and environment characteristics

The lunar surface is covered by a layer of loose, fragmented material called regolith dispersed over a solid bedrock. It is composed of fine dust formed over billions of years due to constant impacts from meteorites and solar wind radiation and scarce rocks. [14], [15] With most particles ranging between 45-100 μm in length on average [16] that takes on sharp and angular, abrasive shapes[17], regolith dust can easily infiltrate mechanisms and electronic equipment. In mechanical systems, the fine dust can lodge in sliding and rotating components like bearings and hinges, significantly increasing friction between parts. This added friction can lead to mechanical stress, hinder smooth movement, and potentially cause failure in affected parts.[18].

The general ruggedness of the moon's surface greatly varies with the chosen landing site [19]. From heavily cratered regions characterised by steep mountains (lunar *terrae*) to vast, relatively flat plains formed by ancient volcanic activity (lunar *mariae*), it is crucial to have an approximate knowledge of the landing region prior to seeking topographical metrics.

Under the assumption that the landing site will be topographically similar to previous successfully historic landing sites, several bounds for topographic parameters can be found to precisely characterise the surface that has to be treated, using previous landings data. [20]. Figure 2.2 shows some of the Apollo missions landing sites that were considered in [20] to gather topographic data, while Table 2.1 quantifies the expected bounds for different topographical parameters on landing sites.

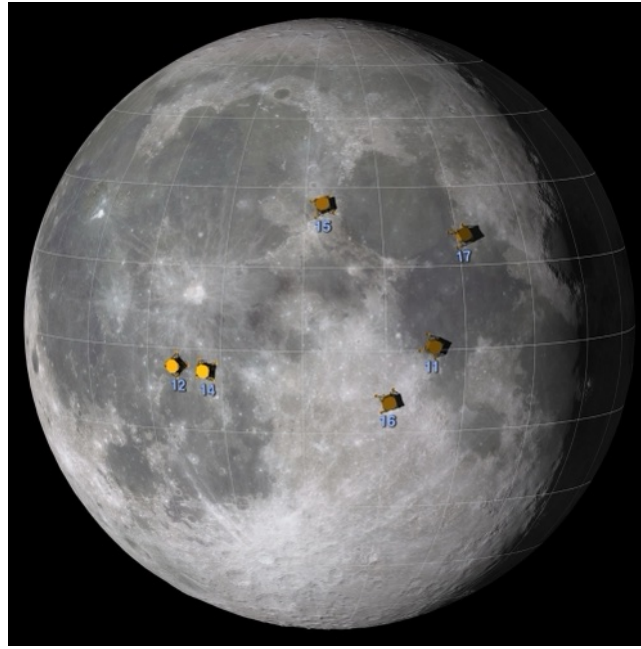


Figure 2.2: Examples of successful Apollo mission landing sites. Data from these missions, including regolith samples and topographical measurements, offers valuable insights for predicting surface conditions at future landing sites. Most Apollo sites are situated in the lunar *maria* regions (shown in dark gray areas).

Parameter	Value	Source
Maximum slope	8.39 [°]	[20], [21]
Maximum Terrain Ruggedness Index	0.23 [<i>m</i>]	[20], [21]
Average Rock Abundance	1.5 [%]	[20]

Table 2.1: Bounds for topographical metrics based on previous NASA's landing sites data.

Where :

- **Maximum Slope** Is the maximum gradient of the terrain. Derived using the Horn and 4-corners algorithms over a 200m diameter circle around the landing sites.
- **Terrain Ruggedness Index (TRI)** Quantifies surface variability by measuring elevation differences over the same circle area. Derived using the Wilson algorithm.
- **Rock Abundance** Measures the proportion of rock coverage in a given area. Derived using manual geolocation over surface temperature data on each historic landing site.

From these metrics, we can conclude that the surface to treat will most likely be relatively flat, lacking landmarks (eg. rocky hills, buildings) and features (eg. segments, corners, geometrical shapes) as depicted in Figure 2.3.

This unstructured environment makes the processing of point cloud data tedious and overall less accurate and efficient. Meanwhile, the flattening process requires an accurate dense mapping of the surface in order to detect small variations in the treated surface.

Additionally, the Moon's intense lighting variations add complexity to operations. Depending on the surface location and time of the lunar day, illumination ranges from harsh direct sunlight—amplified by the reflective regolith—to complete darkness. Near the poles, certain regions remain permanently shadowed (Permanently Shadowed Regions), making visual cues for terrain assessment even more challenging to discern [22], [23].

These considerations taken together represent part of the localization challenges.

The lunar soil consists of varying-density regolith layers over a solid bedrock. Due to the Moon's low gravity and the abrasive nature of regolith, the top few centimeters are loosely packed and compress under even mild pressure as shown in Figure 2.4. This compressibility introduces uncertainty in navigation, as the regolith compacts beneath the wheels. Furthermore, if the underlying bedrock is uneven, compaction may occur laterally, causing unpredictable wheel slip. The low gravity also reduces traction, making it more challenging for wheels to grip the surface effectively. Rocks of varying sizes and shapes scattered across the regolith layer and may roll or shift under the wheels when driven over, contributing to instability and unpredictability in movement [24], [25].

These factors together underscore the inherent navigation challenges posed by the lunar surface, where loose regolith layers, low gravity, and scattered rocks make maintaining stability and traction especially difficult.

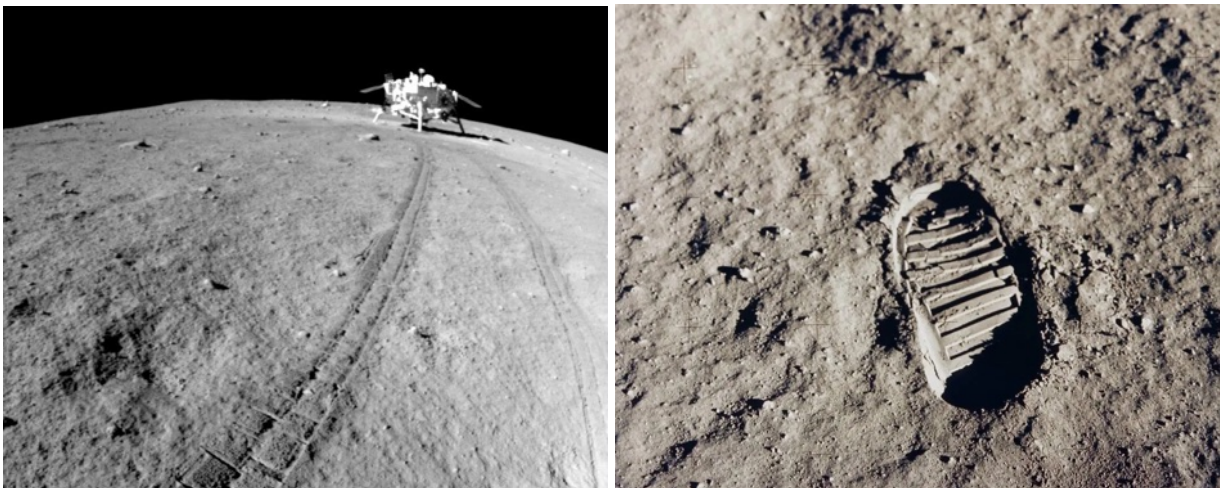


Figure 2.3: Areas that are potential candidates for landings and habitat building on the moon have sparse, unstructured environment that do not facilitate localization because of their lack of landmarks. Picture taken by the CNSA Chang'e 3 Yutu's rover [26].

Figure 2.4: Close-up of regolith top layer, showing compaction even with low amounts of pressure. The unevenness of the underlying bedrock can be seen after the compaction of the top layer by the boot prints. Picture from Buzz Aldrin's bootprint on NASA's Apollo 11 mission [27].

2.2.2 Surface preparation standards

In order to build lunar bases and landing pads, the treated surface has to meet requirements in terms of minimal area size, flatness and compaction ratio.

According to F. Ruess et al., the ideal floor area required for both human operations and storage for lunar missions is approximately 35 m² per crew member [28].

Considering a four members crew aligning with NASA's upcoming Artemis 4 mission requirements, the total needed surface to treat amounts to 140 m² [29].

From [30], we find surface requirements for the topographical metrics described earlier. The values for maximum slope and maximum Terrain Ruggedness Index (TRI) in this study were selected based on engineering requirements for safe lander operations, mobility, and habitat stability. The maximum allowable slope for habitats and landing pads is set at 5°, as this ensures sufficient stability for both applications.

For terrain variability, the target TRI is set to 0.2 meters, minimizing unevenness that could compromise habitat stability and mobility.

Furthermore, rocks and debris are considered a potential hazard for construction and preparation of landing sites. This includes rocks that are partly or entirely covered with the top few centimeters of regolith dust. These need to be removed from the designated areas to ensure safe and reliable landing pads and habitats construction [31]. Table 2.2 sums up the desired characteristics of the treated surface.

Parameter	Value	Source
Maximum slope	5 [°]	[30]
Maximum Terrain Ruggedness Index	0.2 [<i>m</i>]	[30]
Average Rock Abundance	0 [%]	[31]

Table 2.2: Target topographical values after surface treating

By comparing these values to the expected ones for untreated areas in Table 2.1, we observe that the main challenges are reducing the maximum slope from 8% to 5% and clearing rocks from the area.

2.2.3 Restricted sensor usage in lunar environment

Certain characteristics of the Moon prevent the use of some sensors commonly employed on Earth. GPS, widely used on Earth for centimeter-level accurate positioning, is currently impractical on the Moon due to the absence of a dedicated satellite network. Although leveraging Earth's GNSS network for lunar localization is technically possible, the resulting accuracy of 200–300 meters is insufficient for our application [32]. However, ESA's Moonlight program aims to establish the first satellite constellation dedicated to lunar telecommunication and navigation [33]. This network will enable precise autonomous landings, surface mobility, and efficient communication between Earth and the Moon. While unavailable for now, this system should be considered for future iterations of the project, offering enhanced localization and movement capabilities.

Magnetometers, valued for providing absolute heading on Earth, are ineffective on the Moon due to its weak magnetic field [34]. This limitation eliminates them as a viable option for localization or pose estimation. Consequently, the lack of global localization methods necessitates a local mapping approach to navigate the lunar environment effectively [35].

Ultrasonic sensors, commonly used for short-range detection, are also unsuitable for lunar operations. These sensors rely on sound waves, which require an atmosphere to propagate. The Moon's lack of an atmosphere renders this technology unusable.

Another critical consideration is the extreme temperature fluctuations on the Moon's surface. Apollo 12 landing site data shows temperatures ranging from -187°C to 124°C, with similar conditions at other sites [36]. Sensors must remain accurate and reliable within this range. This can be achieved by

designing temperature-resistant sensors or by housing them in a temperature-controlled environment to ensure they consistently operate within their required range [37].

2.2.4 Weight of equipment

Transporting equipment on the lunar surface incurs significant costs, primarily determined by the weight of the payload. Consequently, a key challenge is to minimize the total weight of the robot [38].

2.2.5 Regolith manipulation tools

Flattening a designated surface implies having to manipulate regolith by getting rid of the excess matter and filling cavities, as well as carrying solid matter such as rocks and pebbles out of the area. Additionally, it has been considered to build protective berms around the treated area. In order to do these tasks, one or a set of embedded tools is needed for each step of the surface preparation process. The tools should not hinder the proper functioning of the sensors and wheel motors during their operation.

Rock and debris clean-up The initial phase of surface preparation involves clearing rocks and debris from the area. Orbital images combined with state-of-the-art algorithms can map the sites of interest with a meter to sub-meter resolution at best, implying a certain level of uncertainty regarding characteristics of rocks and debris that are actually on site. As the tool traverses the surface without prior knowledge of rock locations, it may encounter rocks that exceed its handling capacity in terms of size or weight [39].

Excavation tool With a density of approximately 1.5 grams per cubic centimeter, regolith's density is comparable to that of dry sand. However accounting for the Moon's low gravity ($\frac{1}{6}$ th of earth's gravity approximately), regolith doesn't necessarily require the same kind of Earth's heavy-duty systems to be pushed and manipulated. Due to its volatile nature and abrasive properties for nearby equipment, it must be carefully handled and contained to minimize the amount dispersed into the surrounding environment.[40].

Specifically, the tool should easily penetrate the top few centimeters of the regolith layer, push the material while maintaining a consistent depth, and then raise its tool above the ground when not in use, allowing it to move freely without dragging or resistance. During the raking of the top regolith layer, the tool may encounter rocks and debris that were missed in the clean-up phase due to being buried too deep. These remaining solids can interfere with the raking process and potentially damage the tool.

Building habitats on the moon requires managing regolith *ejecta*. When spacecrafts take off or land, their engines stir up regolith particles, creating a high-speed dust cloud that can damage nearby habitats (see Figure 2.5). A common protective measure is constructing berms around the habitat zone to divert regolith away from buildings as shown in Figure 2.6. Research by J. E. Lane et al. indicates that such berms become effective at heights of around 2 meters. [41] Building tall structures with a loose material is not an easy task as the berm can collapse if the base is not stable enough. Moreover, the tool should always be able to add more material until the desired height is achieved without compromising the stability of the berm.

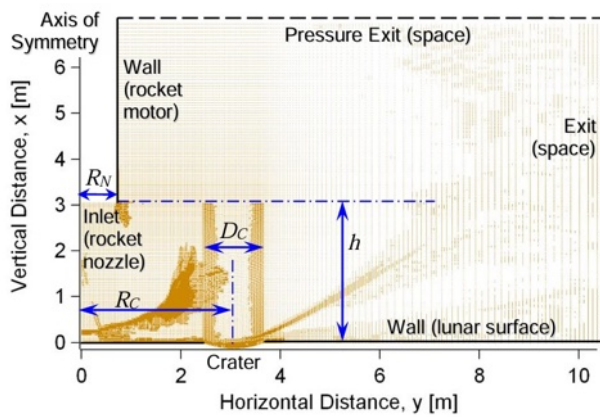
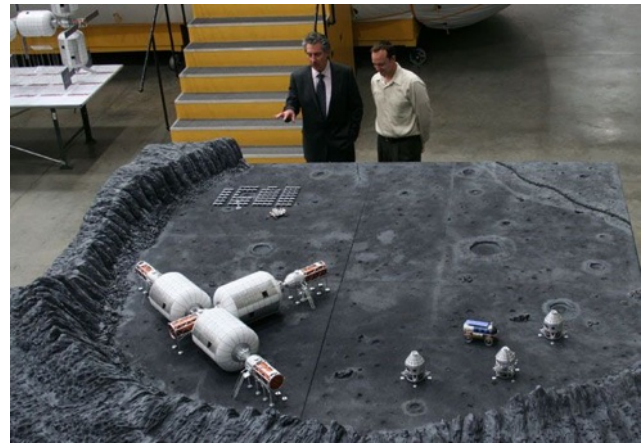


Figure 2.5: CFG simulation of ejecta trajectories due to a rocket nozzle hot gas jet. Particles that start at 10 cm above the surface can be ejected with angles of up to 15 degrees. From [41].



Compaction tool Once flattened and cleared of rocks, the surface needs to be compacted in order to reach a density that can withstand the robot's weight, habitat structures or rocket's weight without further compressing [42].

2.3 Related works and existing advancements

These challenges are actively being addressed, as both complete and partial solutions are being developed and researched. Regardless of many of these solutions being tentative, they present an important stepping stone for state-of-the-art development. Thus, in order to accommodate a system which must address these, existing technologies should be investigated and evaluated.

NASA has devoted considerable attention to the field of regolith manipulation and the development of autonomous robots designed for operation on the Moon.

A number of studies attempt at addressing these challenges collectively, such as the Jet Propulsion Laboratory's (JPL) project; Cooperative Autonomous Distributed Robotic Exploration (CADRE), which consists of three rovers aimed for lunar exploration in 2025. The aim of this project is to test the efficiency applying swarm robotics on the Moon. The rovers in question are depicted in Figure 2.7 and can be seen fitted with solar panneling and is designed to be lightweight. [43]

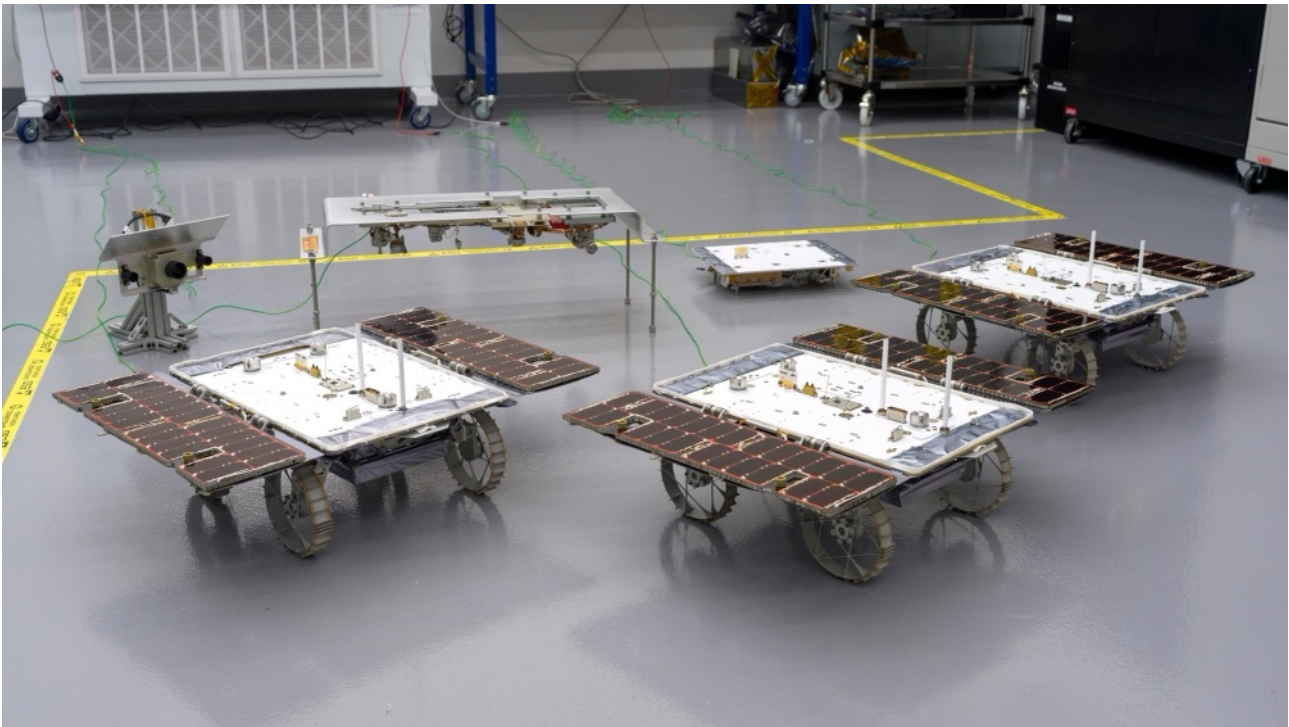


Figure 2.7: CADRE rover displayed in a clean room [44]

The rovers are equipped with two stereo cameras, IMUs, and multistatic ground-penetrating radar for subsurface research over a 400 m^2 area. A secondary mission objective is to demonstrate multi-agent autonomous systems on the Moon. The CADRE project uses MEXEC for path planning, RGB-D cameras for obstacle avoidance, and a leader-based architecture to distribute tasks and assign scanning areas to each robot. The rovers will operate with minimal human intervention. The system is designed with multiple of the constraints defined in Section 2.2; it provides a dense map, skid-steering for slippage on the surface and employs measures against the hazardous temperatures. Furthermore, it can be deployed in a swarm configuration, thereby potentially increasing efficiency. However, it accomplishes only tasks related to exploration and scientific discovery and not excavation, compaction or clean-up. [43]

Another project, The Autonomous Site Preparation: Excavation, Compaction, and Testing (ASPECT) project [13], developed by the Colorado School of Mines in collaboration with NASA, aims to build a fully autonomous lunar rover for preparing landing sites for future missions. The rover, illustrated in Figure 2.8, features an RGB-D camera and LiDAR for environmental sensing, with navigation handled by an Extended Kalman Filter. Although primarily designed for landing site preparation, it is assumed that the system could be adapted for construction tasks as well. The rover is equipped with a custom blade for moving regolith and a specialized compaction tool, both developed for this mission. As the system and hardware was specifically designed for the same task as this project, it is designed to accomplish all the relevant tasks. It also provides a high density map, wheels designed for lunar exploration and localisation in sparse environments. Furthermore, the system of ASPECT is designed to perform all the tasks presented in Section 2.1. [13]

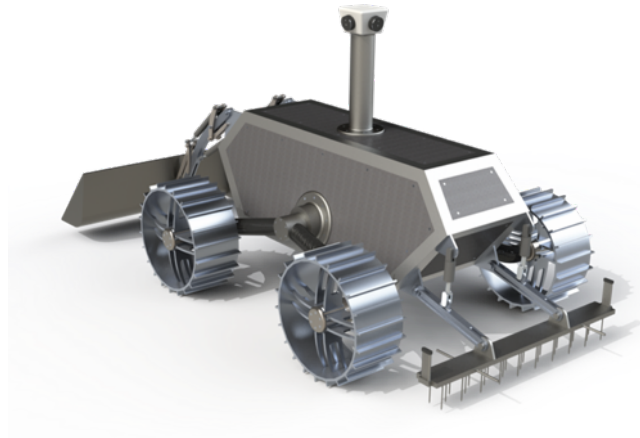


Figure 2.8: Depiction of the conceptual model of the ASPECT, taken from [45]

NASA's research facility, Swamp Works, specializing in granular mechanics and regolith operations, has also developed a tentative solution, named Regolith Advanced Surface Systems Operations Robot 2.0 (RASSOR) for manipulating, excavating, hauling, and dumping regolith on extraterrestrial surfaces. While the robot is designed to autonomously map its environment using stored hazard maps and stereo vision for landmark identification, the project's main focus was testing the robot's bucket capabilities. In later stages, the system's overall functionality was further evaluated. The second version of RASSOR uses stereo cameras for obstacle detection and integrates IMU data through sensor fusion for navigation. The system addresses some of the challenges specified, Dense mapping, Slippage, autonomous operation and sparse localisation. However, it is required to return to the lander for charging and during lunar nighttime. Furthermore, it is only capable of performing two of the five task categories, specifically, exploration and excavation. [46], [47]

Concluding upon these findings, the advantages and disadvantages, in relation to the challenges presented, can be seen in Table 2.3.

Challenges	CADRE	ASPECT	RASSOR
Dense Map	✓	✓	X
Temp. Resilient	✓	?	X
Addresses Slippage	✓	✓	✓
Autonomous	✓	✓	✓
Sparse localisation	✓	✓	✓
Swarm Capable	✓	X	X
Weight	< 10 kg	< 83 kg	66 kg

Table 2.3: Summary of the challenges that the proposed full solutions can at this point accomplish, based on the descriptions above. The individual symbols mean: ✓ = Yes, X = No, ? = Not specified

Similarly for the task capabilities of each of these solutions can be see in Table 2.4

Challenges	CADRE	ASPECT	RASSOR
Exploration	✓	✓	✓
Excavation	X	✓	✓

Challenges	CADRE	ASPECT	RASSOR
Compaction	X	✓	X
Dust mitigation	X	✓	X
Clean-up	X	✓	X

Table 2.4: Summary of the challenges that the proposed full solutions can at this point accomplish, based on the descriptions above. The individual symbols mean: ✓ = Yes, X = No

As none of these project fully cover all the challenges, nor the tasks required, a combination of the ideas presented in these could provide a stepping stone for further developement and design in this project. These projects offer comprehensive solutions to critical challenges in lunar exploration, such as autonomous site preparation, regolith manipulation, and advanced navigation. While the above projects focus on a complete solution that address all of these challenges, individual challenges are also being actively developed.

2.3.1 Localisation and mapping

As discussed in Section 2.2.1, the lunar surface is relatively unstructured and sparse, limiting the effectiveness of traditional localisation and mapping methods that rely on numerous distributed keyframes or landmarks [48]. Therefore, robust alternatives are essential for this project and future lunar exploration and construction. In cases where surface manipulation is required, a dense map becomes necessary to monitor outcomes, efficiently locate areas of interest, correct errors, and track metrics against specific requirements. To meet these needs, the system must provide both dense mapping and localisation in sparse environments.

Visual SLAM has emerged as a promising approach for extraterrestrial mapping. Hong et al. [49] propose a Stereo Parallel Tracking and Mapping (S-PTAM) system with self-supervised deep learning, which supports dense mapping and localisation in sparse environments. However, this method is vulnerable to illumination changes, particularly in permanently shadowed regions (PSRs) on the Moon. [49]

To address these low-light conditions, Park et al. [50] augment Visual SLAM with low-light image enhancement (LLIE), improving keyframe detection and mapping accuracy under low-light conditions. Results from this approach, depicted in Figure 2.9 and Figure 2.10, indicate a robust localisation solution for sparse and low-light environments, that also provides a dense map. [50]

Alternatively, Giubilato et al. [51] introduce GPGM-SLAM, which uses Gaussian Process Gradient Maps (GPGMaps) derived from terrain elevation gradients. This method supports loop closure detection and localisation in feature-sparse areas but does not produce a dense map. [51]

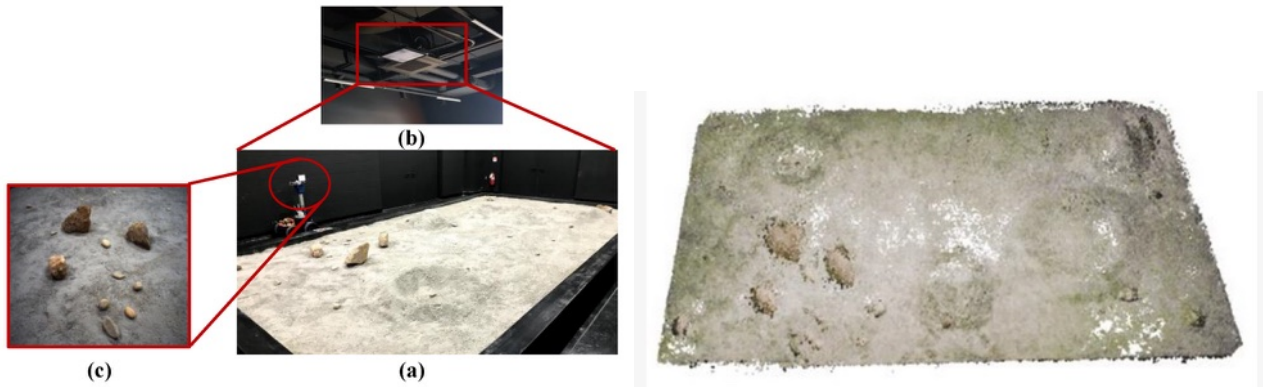


Figure 2.9: Depiction of the testbed used in study by **Figure 2.10:** Depiction of the coloured pointcloud re-Park et al. **a)** A testbed filled with regolith simulant sulting from the system implemented by Jae-Min Park (KLS-1), **b)** Adjustable light level LED lamp, **c)** Image et al. [50] of rocks added to the test bed. [50]

In PSRs, Tong et al. [52] and Du et al. [53] suggest active sensing approaches with LiDAR and laser range finders, along with point-cloud feature matching. Tong et al. propose a 2.5D SLAM approach with stop-scan-go to reduce motion blur and achieve consistent dense mapping, complemented by odometry. Du et al., on the other hand, use LOAM with back-end optimization to reduce computational load, though it does not support dense mapping. The approach used by Tong et al. achieves a position error of 0.35 m, orientation error of 2.11° , and map error of 0.04 m, supporting effective localisation in sparse environments. [52], [53]

Lee et al. [54] take a different approach using horizon masks created from NASA's high-resolution digital elevation map (DEM) data from the Lunar Orbiter Laser Altimeter (LOLA). This method matches stored horizon masks to observed horizons for localisation, see Figure 2.11, achieving meter-level localisation accuracy in simulation when the instrument tilt is within bounds. [54]

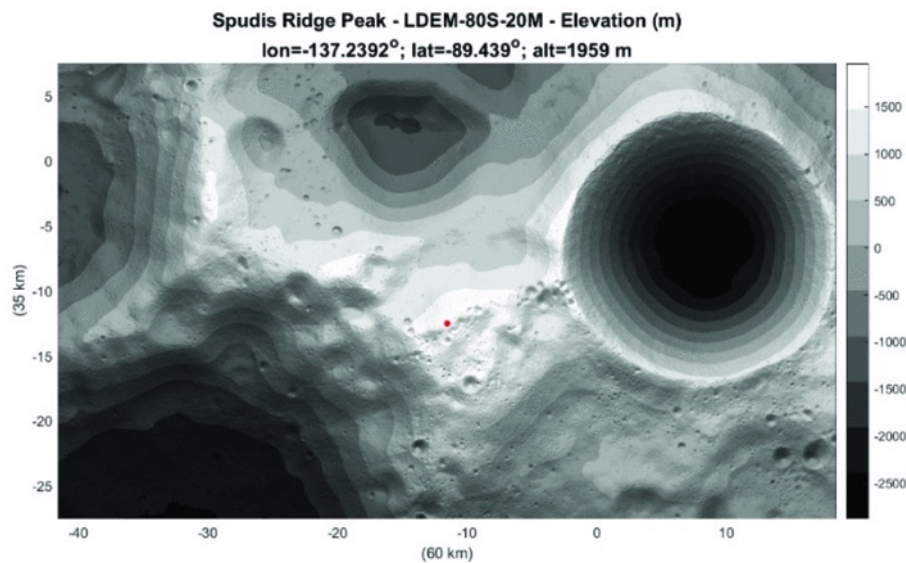


Figure 2.11: Elevation map of the Lunar South Pole. Landmarks: Spudis Ridge (red dot) and the Shackleton Crater (right) [54]

Recent studies have also explored multi-robot approaches to localisation. For example, Bahraini et al. [55] introduce cooperative visual odometry (CVO), using overlapping fields of view between robots to improve localisation through loop closures. Although CVO does not generate dense maps, it enhances localisation accuracy through shared observations. Lastly, van der Meer et al. [56] present REALMS, a decentralized SLAM system for multi-robot extraterrestrial applications, using an RGB-D camera and IMU to address localisation drift due to wheel slippage. REALMS successfully achieved 19.4% terrain coverage in ESA's Space Resources Challenge, it does however not provide a dense map.

Concluding upon these findings, the advantages and disadvantages, in relation to the challenges presented, can be seen in Table 2.5. In addition, the capability of the proposed solution to be illumination invariant based on these findings was added.

Author	Proposition	Dense map	Sparse localisation	Illumination invariant
Hong et al.	S-PTAM and Deep learning	✓	✓	X
Park et al.	VSLAM and LLIE	✓	✓	✓
Giubilato et al.	GPGM-SLAM	X	✓	X
Tong et al.	2.5D SLAM	✓	✓	✓
Du et al.	LOAM	X	✓	X
Lee et al.	Horizon masks	X	✓	X
Bahraini et al.	CVO	X	✓	X
van der Meer et al.	REALMS	X	✓	?

Table 2.5: Summary of the capabilities of the different approaches presented for localisation and mapping. Symbols: ✓ = Yes, X = No, ? = Not specified

As can be seen only VSLAM enhanced with LLIE, as proposed by park et al [50], fulfils all the criteria. However, it does not present an entire solution, and is therefore only considered a direction for further development.

2.3.2 Navigation and Path planning

As with localisation and mapping, navigating and planning paths on the lunar surface present unique challenges, including uneven terrain, the need for 3D pathfinding. It is further complicated by the absence of external navigation infrastructure like GPS, the hazardous terrain and the instability of the regolith. The regolith can interfere not only with sensory hardware, but also with mobility, as described in Section 2.2.1. Consequently, the system must efficiently plan its path, effectively avoid obstacles and conflicts, and securely navigate in 3D space; capabilities essential for precise surface manipulation. Furthermore, since human intervention is extremely limited on the lunar surface, the system must be energy efficient.

An approach presented by Tanaka and Malki [57] proposes a deep learning framework for lunar rover path planning that considers environmental and resource constraints, including thermal and energy efficiency. By incorporating reinforcement learning, the method adapts path planning to handle both time-variant and path-dependent constraints, such as lunar terrain and the rover's internal power and thermal status. While effective for efficient, autonomous navigation, the framework does not specify

if it supports 3D navigation. [57]

Similarly to localisation, multi-robot propositions are emerging for extraterrestrial applications, which also requires the system to handle path-conflicts. Two papers by revolving around the Ant Colony Algorithm (ACO) both relying on centralised applications, the first being a study by Zhu et al. [58] propose a centralized approach for multi-robot lunar navigation using Ant Colony Optimization (ACO) with adaptive potential fields to manage dynamic and stationary obstacles. This method achieves a success rate of over 60%, effectively handling path conflicts, though it lacks consideration for 3D pathfinding, temperature, or energy efficiency. [58]

A study by Lu et al. [59] alternatively integrate Artificial Potential Fields (APF) with Deep Reinforcement Learning (DRL) to enhance ACO-based path planning. APF provides obstacle avoidance, while DRL improves path efficiency in complex lunar terrain, showing stable navigation and reduced computation, although it has issues with trapping and 3D pathfinding, temperature, and energy efficiency remain unaddressed. [59]

Lastly, Feng et al. [60] presents a deep learning-based approach for multi-agent path planning on the lunar surface, combining an improved Conflict-Based Search (CBS) algorithm with a Deep Q Network (DQN) to efficiently resolve navigation conflicts among lunar robots. DQN prioritizes conflicts, enabling CBS to generate collision-free paths quickly, resulting in high efficiency and reliability. However, this method does not address energy or thermal efficiency, nor does it support 3D pathfinding. [60]

Concluding upon these findings, the advantages and disadvantages, can be seen in Table 2.6.

Author	Proposition	Conflict avoidance	Energy efficient	Temp. efficient	3D pathfinding
Tanaka and Malki	Res. DPF	X	✓	✓	?
Zhu et al.	ACO	✓	X	X	?
Lu et al.	APF and DRL	✓	X	X	?
Feng et al.	CBS and DQN	✓	X	X	?

Table 2.6: Summary of the capabilities of the different approaches presented for navigation. Symbols: ✓ = Yes, X = No, ? = Not specified

In conclusion to the above mentioned propositions, none of the provide a complete solution, nor do any of them specify the ability to pathplan or navigate in 3D environments. However, for the development of this project, a solution incorporating all these would be desirable.

2.3.3 Robot- and tool design

Although the above research provide tentative solutions to some of the challenges presented in section 2.2. The system would still require the tools to perform the tasks specified in section 2.1. Thus, an analysis of existing designs of these tools and their capabilities is crucial for further development of a full system.

Mueller et al. [61] propose Attachment Node for Construction & Excavation (LANCE), a proof of concept excavation and grading tool. The tool is composed of an Aluminium support frame and mount plate, a composite moldboard and a hardened steel cutting edge and backer plate, as can be seen

in Figure 2.12. [61]

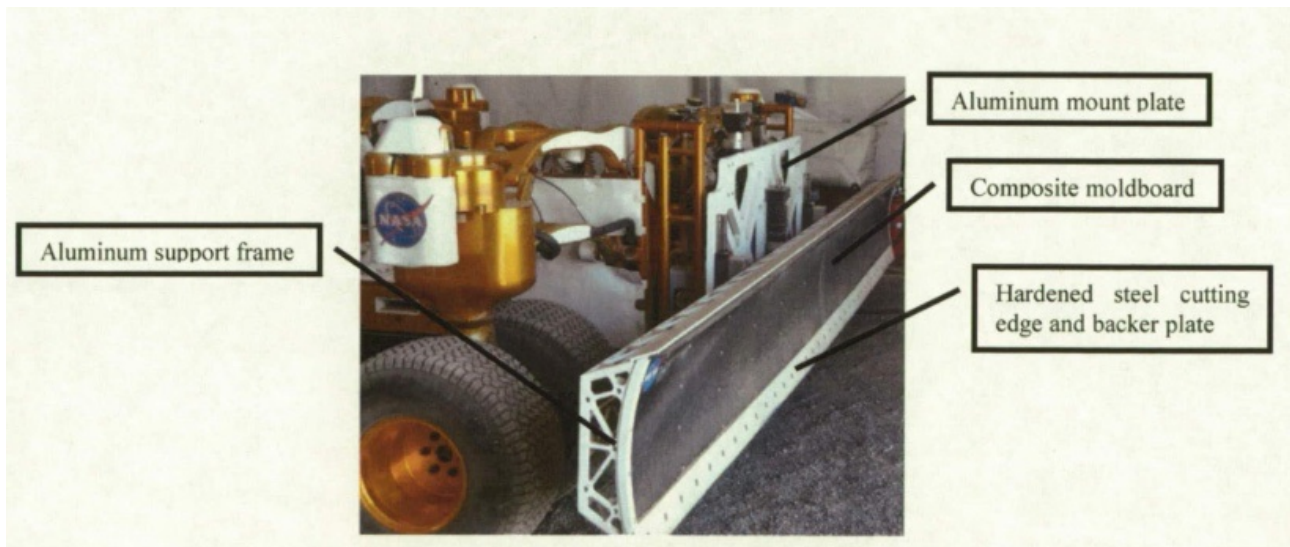


Figure 2.12: Depiction of the LANCE mounted on a Chariot, taken from [61]

The capabilities of the tool was tested multiple times at varying locations, which showed that it had the capabilities of both excavation and grading (part of compaction). Furthermore, the material and structure of this tool is 70% lighter than an equivalent hardened steel tool. [61]

A Earth alternative is presented in a study by Selech et al. [62]. They propose a design for a bulldozer blade as an attachment for a compaction drum roller, aimed at expanding the roller's functionality in construction tasks. The proposed C-type blade allows for tasks such as ground leveling and trench filling. Constructed from RAEX 400 steel, the blade is optimized for strength and durability, as validated by FEM analysis, ensuring it can withstand operational stresses without deformation or weld failure, but also adding weight to the design. [62]

These designs, could also partly be used for clean up, pushing away loose rocks and debris. However, they might not be suitable for moving objects that are stuck in more compacted regolith. Traction will also be an issue in this case, and therefore they are only considered to partly fulfil this purpose.

However, lunar regolith often includes larger, dense clumps rather than loose sand. A paper by Iai et al. [63] presents a blade designed specifically to improve excavation efficiency in such conditions. This design pairs a ripper with a wide blade that follows behind, allowing the ripper to first break up compacted regolith, reducing the total excavation energy required, as seen in Figure 2.13.

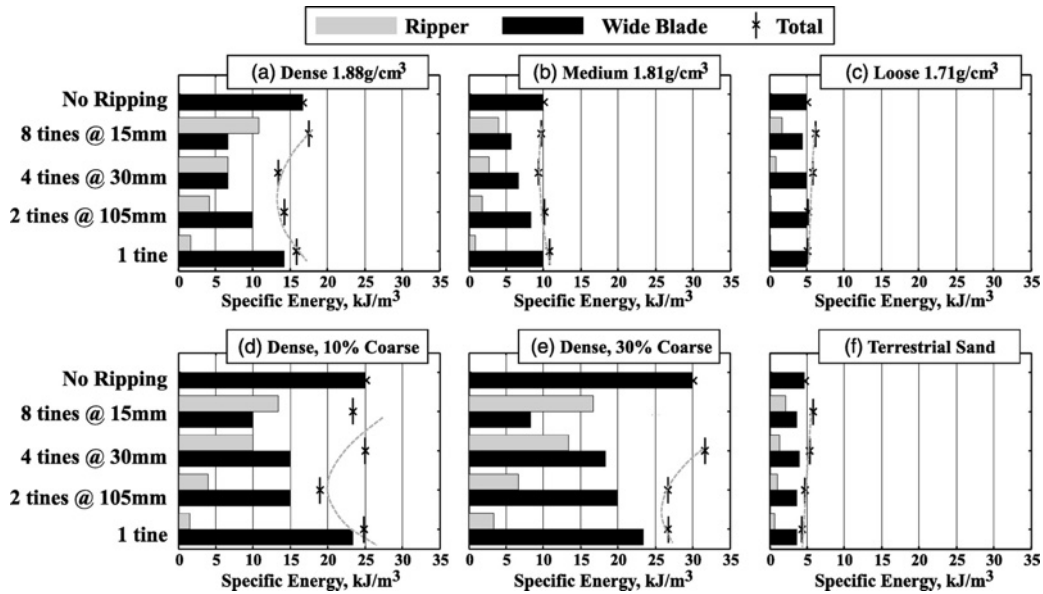


Figure 2.13: Results taken from [63], showcasing the required energy for excavation with no ripping and with various amounts of tines and distance between tines, the dashed lines are trends.

Tests show that pre-ripping in high-density regolith can lower excavation resistance significantly, making the subsequent blading process more efficient. This approach is especially suited for coarse-grain regolith and supports In-Situ Resource Utilization (ISRU) by optimizing the excavator's performance on the Moon's surface. Although this tool does not accomplish any of the required tasks fully, it introduces energy efficiency into excavation, and is therefore considered partly fulfilled in this regard. [63]

Another approach to excavation involves moving lunar regolith by loading it into containers or transporting it one scoop at a time. A paper by Zacny et al. [64] proposes a lunar backhoe designed to excavate and transport regolith effectively in the Moon's low-gravity environment. The honeybee scoop utilizes a percussive digging mechanism, which minimizes the reaction forces required for digging, allowing for efficient scooping of compacted regolith. The percussive motion enables the scoop to penetrate hard-packed surfaces while also shaking free any clinging particles, making it highly suitable for ISRU tasks like trenching, habitat construction, and berm building. [64]

A similar proposition by Mueller et al. [65], known as the Vibratory Impacting Percussive Excavator for Regolith (VIPER), was also designed to operate with a percussive mechanism, here attached to a pivoting bucket. This tool uses impact energies ranging from 13.6J to 30.5J at frequencies up to 700 beats per minute (BPM). The tool achieves a notable reduction in excavation load, up to 50%, enabling a lighter excavator design suitable for space missions, particularly in harsh, compacted regolith conditions. [65].

A slightly different approach is presented on the RASSOR by mueller et al [66], as mentioned earlier, it relies on a bucket system, designed specifically for excavation in low-gravity environments, depicted in Figure 2.14. The bucket system consists of two counter-rotating drums that serve both to dig and to transport regolith. Each drum is equipped with scoops along its edge, which allows it to collect soil as it rotates forward. For soil deposition, the drum rotates in the opposite direction, releasing the collected material. This counter-rotation design not only enables effective excavation but also stabilizes the robot, reducing the need for heavy weights, which is crucial in lunar and Martian environments. The RASSOR bucket system is optimized for continuous excavation, providing

an efficient, lightweight solution for resource collection. It does however, not support picking up or moving larger rocks. [66]



Figure 2.14: An illustration of the RASSOR, with the bucket system, taken from [47]

A terran alternative by Saldaña-Robles et al. [67] presents a design with emphasizes weight reduction and structural resilience. Using FEA, RSM, and ANN, they optimized components like the bucket, arm, and boom, achieving a 24.8% weight decrease without compromising strength. Constructed from ASTM-A572 steel, the backhoe is reinforced to withstand high breakout forces, making it ideal for demanding agricultural tasks such as trenching and excavation. [67]

Lastly, being able to compact the regolith to the specified degree requires alternatives, traction and weight has a lower impact on the lunar surface. A proposition was presented on the ASPECT by Dreyer et al. [13] suggests vibratory method, where a set of needles is inserted into the regolith, through a plate. Then vibrating the these needles aligns the grains of the regolith making it more compact, as illustrated in Figure 2.15. [13]

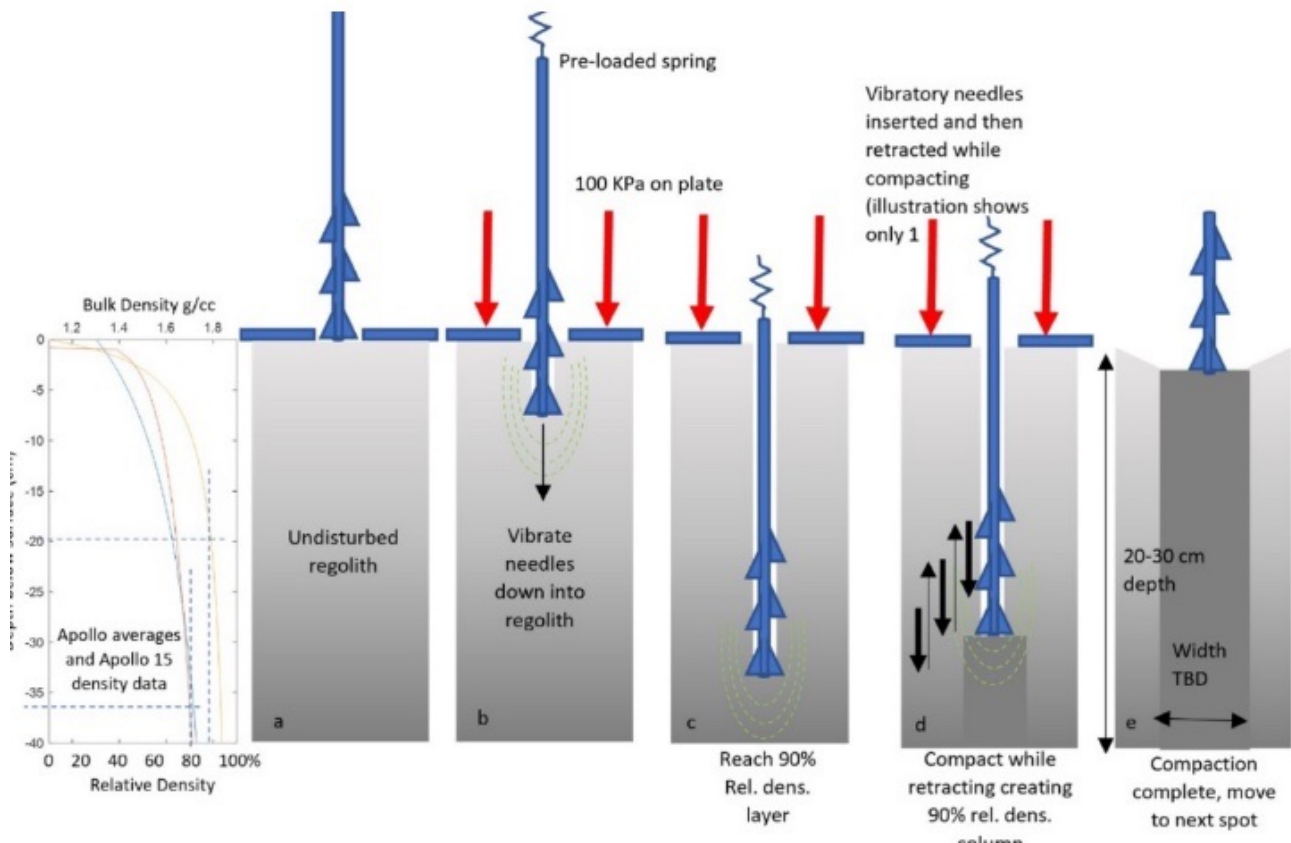


Figure 2.15: Illustration of the concept for the compaction tool designed for the ASPECT. **a)** The plate and needles are aligned with the surface. **b)** A 100 kPa is applied to the plate and the needle is inserted into the regolith. **c)** The needles vibrate to align the grain and increase the relative density. **d)** During retraction and extension the needles continue to vibrate to ensure a distributed relative density. **e)** the needles are retracted and a more compact regolith area is left behind. Taken from [13]

This concept, however only achieves a relative density of up to $\sim 65\%$, and cannot be used for other tasks. [13]

A set of terrestrial alternatives for vibratory compaction technologies is presented in a paper by Pankrath et al. [68]. The study describes several dynamic compaction methods, including Dynamic Intensive Compaction, Rapid Impact Compaction, and the BOMAG Impact Roller. These systems employ high-frequency, transient vibrations to achieve deep soil compaction efficiently. Materials such as high-strength steels are used for durability, and the equipment often features polygon-shaped or rigid compaction feet to enhance ground interaction. Each technology is optimized for specific applications and depth requirements, with numerical simulations and physical tests confirming their effectiveness in achieving uniform, high-quality compaction. However, they again cannot perform other tasks.

Concluding upon these findings, the different tasks that the individual tools can accomplish are summarised in Table 2.7 along with a empirical evaluation of whether they are designed to be lightweight enough for lunar operation.

Tool	Lightweight	Excavation	Compaction	Clean up
LANCE	✓	✓	X	P
S-Blade	X	✓	X	P
Ripping	✓	P	X	X
Honeybee scoop	✓	✓	X	✓
VIPER	✓	✓	X	✓
RASSOR Bucket	✓	✓	X	X
Terran backhoe	X	✓	X	✓
ASPECT pins	✓	X	✓	X
Ter vibration	X	X	✓	X

Table 2.7: Summary of the capabilities of the different tool designs. Symbols: ✓ = Yes, X = No, P = Partly

Based on these findings it is evident, that one or more multifunctional tools are required for the purpose of this project. Furthermore, the tools should be a combination or compromise of weight, strength and functionality.

2.3.4 Dust mitigation

Any kind of robotic operations on the lunar surface inevitably agitates the soil, dispersing the fine regolith dust in the air. This is due to the volatile nature of the regolith and the moon's weak gravitational field.

To prevent regolith from entering sensitive parts of lunar equipment, several methods have been proposed.

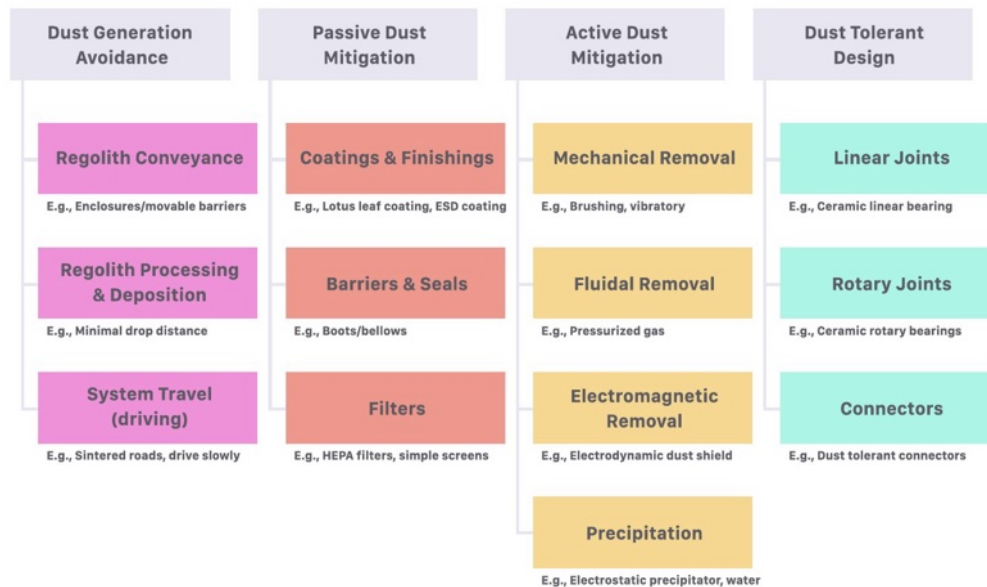


Figure 2.16: Overview of the different dust mitigation processes that are currently studied.

Ginés-Palomares et al. proposed to melt regolith samples into tiles using, in order to lay pavement.[69] E. Cardiff et al. suggested using Fresnel lenses to focus solar energy directly onto the lunar soil, creating a thin layer of melted regolith on top of the loose dust layer[70] while K. W. Farries et al. are proposing the same principle using embedded lasers [71].

2.3.5 Conclusion

Based on these studies, a solution incorporating localisation based on multiple factors is preferred as it provides a more robust solution to a multitude of situations. However, for the purpose of localisation and mapping a solution incorporating SLAM is advantageous since it is adaptable to multiple scenarios. Furthermore, SLAM is often built around robust solutions and is available for both single and multi-robot applications. For navigation a multitude of options exist and only a few are mentioned here, it is however clear that the solution must incorporate adaptations to terrain, energy and temperature efficiency, and path conflict management. For terrain manipulation a range of tools for the specific tasks have been considered, designed for both extra- and terrestrial applications. Based on the analysis of the varying tools is evident that they must be designed with weight, robustness and versatility in mind. Lastly, dust mitigation techniques must be designed and implemented in a resource- and energy efficient way for it to be a viable solution for prolonged autonomous operation on the lunar surface.

3 Problem outline

With the completion of preliminary research and establishment of challenges and related work the final problem statement, and hereafter the requirements and objectives can be outlined. However, as this project is constrained by the scope of a semester, to reduce the scope and fit within this time frame some limitations need to be outlined. These limitations, detailed below, have been carefully considered through the preparation of the report and the research conducted in the previous chapter.

3.1 Delimitations

As described, due to the inherent nature of this project and the scope and resources of general space related projects, this project can not accomplish the tasks of both designing and constructing a rover and implement a control scheme. Thus, these limitations, listed below, are based upon the time-frame available, and the resources and manpower available.

- **Time:** Being constrained to a single semester, this project operates within a finite time frame that necessitates that the conceptualisation and development focus on a limited area and that the development and implementation phases to be completed efficiently. Thus, for the purpose of this project only parts of the system and simple tooling will be designed, implemented and tested.
- **Resources:** Since building a robotic lunar rover is beyond the scope of this project, the development and implementation will be performed on a from KOMPAĚ robotics (Formerly Robosoft) [72], provided by Aalborg University as part of the project.
- **Requirements:** Due to the general lack of specific measurable parameters available, the inherent nature of academic projects related to space exploration and to reduce the scope only objectives will be outlined in this project.

With these limitations in mind, the remainder of the project can be formed, alongside the development of the proposed prototype. The following sections establish a final problem statement and outline the objectives for the project.

3.2 Final problem statement

Thus from the research conducted in chapter 2, it can be concluded that some areas of research can be combined or used for development of a solution for the purpose of this project. Based on this and the requirements posed by NASA in other projects the problem statement is refined to:

”How can a system be designed efficiently for rovers to prepare the lunar surface for construction and landing sites.”

3.3 Objectives

Based on the research conducted, the task of this project can be divided into a set of objectives that needs to be addressed or solved for a full solution. These are as follows:

1. Design and implement a robust interfacing framework that supports integration with a known mobile platform. Ensure the framework is adaptable and scalable to accommodate future modifications or additional components. The framework should facilitate seamless interaction between sensors, actuators, and processing units for efficient system operations.
2. Determine and implement a localization and mapping algorithm capable of operating effectively in sparse and unstructured environments. The algorithm should enable precise localization of the platform and generate a high-resolution map.
3. Develop and implement a navigation scheme or algorithm that utilizes the dense map, localization data, and sensor inputs. The navigation system should guide the platform across the terrain efficiently while ensuring optimal area coverage.
4. Design and construct simple yet effective tooling for the mobile platform to test its surface preparation capabilities. The tools should be sturdy, reliable and lightweight to perform tasks such as flattening or covering surfaces.
5. Test the system in appropriate environments that simulate real-world conditions to evaluate its performance against the outlined requirements. Assess the communication framework, mapping and localization algorithms, navigation scheme, and tooling. Document and analyse results to determine adherence to project goals and identify areas for improvement.

The following chapters reflect the need to fulfil these objectives.

4 Conceptual design

Based on the presented objectives for this project, an idealised design can be conceptualised, albeit being limited to the available hardware at Aalborg University and the specific platform, these considerations are, by the project group, considered the optimal options. The final hardware that was used is specified in Figure 5.4. The following sections assess a variety of options for both SLAM and tool designs.

4.1 SLAM

As discussed in section 2.3 the proposed system should preferably be based on a SLAM system providing a dense map that can be used both for localisation and analyse the environment for e.g. rocks and other debris. Since the system is supposed to be operating in space and this limits the available sensors as discussed in subsection 2.2.3 some sensors are unavailable for usage in extraterrestrial settings. Thus, sensors such as those reliant on Infra-Red, magnetic fields and similar cannot be used in this connection. Consequently, sensor choice is paramount for the success of the system.

This section explores the sensor options and subsequent available SLAM methods for usage in sparse and unstructured environments. The focus is the mapping sensors as accelerometer and gyroscope is considered essential equipment for the purpose of this system.

4.1.1 Sensory elements

To enable the creation of a dense and detailed map, the system requires exteroceptive sensors capable of high-resolution data acquisition and accurate depth or distance measurements. Considering the unique challenges posed by extraterrestrial environments, as discussed in section 2.2, infrared sensors, magnetometers, and similar technologies are excluded from consideration due to their limited reliability in such contexts. Instead, the focus is on cameras and laser-based solutions, since they better fulfil these criteria and can function in these environments.

To enhance environmental understanding and facilitate the differentiation of terrain components such as rocks, debris, and regolith, semantic information is essential. Consequently, sensors that provide RGB (color) data are particularly desirable, as they enable the system to incorporate visual context into the mapping process. Thus the following list contains these types of sensors and suggestions for the sensors, which are analysed in depth hereafter.

- **RGB-D Cameras**
 - ZED 2i (Stereolabs)
 - Intel RealSense D455
 - Flir Blackfly S
- **LiDARs**
 - Velodyne Puck VLP-16 High Res
 - Ouster OS0
 - Hesai PandarXT
 - Luminar Hydra

The evaluation of the listed RGB-D cameras and LiDAR sensors is conducted based on their capabilities in dense mapping, localization, and semantic differentiation under the challenging conditions of the lunar surface. The metrics considered include depth accuracy, range, resolution, robustness to environmental conditions, and suitability for semantic scene analysis.

RGB-D Cameras

The ZED 2i, Intel RealSense D455 and Flir Blackfly S represent advanced RGB-D camera technologies with distinct characteristics. Among these, the ZED 2i stands out for its superior depth range (0.5–20 meters), making it suitable for mid-range dense mapping. Its power consumption is approximately 1.9 W, comparable to the Intel RealSense D455, which consumes 1.5 W. This makes both sensors viable for energy-constrained lunar applications. However, the ZED 2i offers better depth accuracy (1% of distance) compared to the D455's sub-2% accuracy, favoring it for precision mapping. The Flir Blackfly S, consuming around 2 W, lacks integrated depth sensing but provides excellent image quality for applications that require high-resolution imaging in conjunction with stereo setups. [73]–[76]

In summary, the ZED 2i is the most balanced option for standalone dense mapping and localization, while the D455 offers compactness and slightly better energy efficiency. The Flir Blackfly S, while moderately energy-efficient, is more suited for specialized imaging tasks as part of a multi-sensor system. [73]–[76]

Sensor	Max Range(m)	Accuracy	Power (W)	Semantic Capability
ZED 2i	0.3-20 (2.1mm lens) 1.5-35 (4mm lens)	1%	4.5	Moderate
Intel RealSense D455	0.4–6	< 2%	1.5	Low
Flir Blackfly S	N/ A	N/A	2	Moderate (with AI)

In summary, the ZED 2i is the most balanced option for standalone dense mapping and localization, while the D455 offers compactness and slightly better energy efficiency. The Flir Blackfly S, while moderately energy-efficient, is more suited for specialized imaging tasks as part of a multi-sensor system. While these are based on reflected light, they require either sufficient ambient illumination or light from external light sources. [73]–[76]

LiDARs

Lidars however does not require external light sources for operation, the options presented in the earlier list are here evaluated based on the same criteria. The Velodyne Puck VLP-16, Ouster OS0 and Hesai XT32 vary significantly in their capabilities and energy requirements, each offering unique advantages and trade-offs for lunar exploration. For dense mapping and localization, the Velodyne Puck VLP-16 and Ouster OS0 both provide a maximum range of 100 meters, sufficient for small-to medium-scale mapping on the lunar surface. However, the Velodyne Puck is significantly more energy-efficient, consuming only 8 W compared to the Ouster's 14-20 W. This makes the Velodyne a better option for energy-constrained missions, though its angular resolution of 0.2°-0.4° is less precise than the Ouster's 0.1°, potentially limiting its utility for highly detailed maps. [77]–[79]

The Hesai XT32 extends the range to 120 meters and offers comparable angular resolution to the Ouster. However, this improvement comes with a significant energy cost, as it consumes 18 W. For missions prioritizing extended range and dense point clouds, the Hesai sensor is advantageous, but its higher power requirements make it less suitable for smaller robotic platforms with strict energy budgets. [78], [79]

In terms of semantic differentiation, most LiDARs rely on post-processing or external integration with AI-based systems for object classification. The Ouster OS0 can integrate well with semantic processing systems due to its configurable point cloud density and resolution, making it the most versatile choice for missions requiring both detailed mapping and scene understanding.

Sensor	Max Range(m)	Accuracy	Power (W)	Semantic Capability
Velodyne Puck VLP-16 Hi-Res	100	0.1°-0.4°	8	Low
Ouster OS0	100	0.1°	14-20	Moderate
Hesai XT32	120	±1cm	10	Low

Conclusion

For effective mapping and localization, a combination of LiDAR and RGB-D sensor modalities is considered optimal. This approach leverages the strengths of both systems, enabling robust performance across a range of scenarios, including areas with limited lighting or shadowed regions where RGB-D cameras alone might struggle. The integration of multiple ZED 2i cameras, strategically positioned to provide wide coverage at the front and central coverage at the rear, ensures comprehensive visual data acquisition. These cameras are complemented by the Hesai PandarXT-32 LiDAR, which offers adaptive accuracy across varying distances, making it particularly well-suited for precise spatial mapping and localization in dynamic environments. This hybrid setup enhances overall system reliability by addressing the limitations of each individual sensor type, ensuring high-resolution mapping and robust obstacle detection across diverse lunar terrains. Since these are to be used in a SLAM setting, the algorithm that best fits these sensors needs to be determined.

4.1.2 SLAM approach

In order to fuse the outputs from all these sensors and create a virtual map of the environment, wherein the system can estimate ego-motion and localise itself. For this purpose a few select algorithms are considered viable, measured in whether they support multiple sensors, provide a dense map, accuracy and their robustness in sparse and unstructured environments. The following algorithms are considered:

- RTAB-Map
- LIO-SAM
- ORB-SLAM3
- LVI-SAM
- VINS-Fusion

In order to select the optimal algorithm for this application, their capabilities in regards to multi-modal sensor support, dense mapping, accuracy and known success rates. RTAB-MAP (Real-Time Appearance-Based Mapping) by Labbé and Michaud [80] is a versatile algorithm supporting RGB-D cameras, LiDARs and stereo systems. It is a graph-based SLAM algorithm that generates dense maps by fusing data the multiple sensor modalities, thus making it a robust choice for applications requiring comprehensive terrain mapping. However, its reliance on loop closure detection to maintain some

form of global consistency can be a limitation in environments where visual features are sparse, as often seen on the lunar surface. Furthermore, the computational requirements can pose a challenge for real-time applications in resource constrained environments. [80]

Another method is utilising Lidar inertial odometry via smoothing and mapping (LIO-SAM) by Shan et al. [81], which attempts to bridge the gap between LiDAR-centric mapping and visual SLAM by integrating LiDAR and IMU data, with optional extensions for visual data. It is capable of producing dense maps with high spatial precision, which makes it a strong candidate for lunar applications, particular large scale terrain mapping. Since it relies on LiDAR point clouds for robust spatial representation it excels in environments with minimal visual features. It often achieves a localisation accuracy of over 95% in terrestrial tests, with significant robustness in feature-deprived settings. [81]

Another option is utilising ORB-SLAM3 by Campos et al. [82], a visual SLAM, mainly based on Monocular, stereo and RGB-D cameras using both pin-hole and fisheye lens models. While it is highly accurate for visual SLAM it is not adaptable for e.g. LiDARs. It proves very efficient in structured environments with a high amount of visual features, often times attaining an accuracy of 90-95%. However, in unstructured or feature-sparse lunar environments, its reliance on these features potentially limits its effectiveness. Furthermore, in order to obtain dense maps from ORB-SLAM3 it requires external tools and thus does not provide a suitable candidate unless these external tools are added with a limited impact on computational resources. [82]

A final standalone SLAM method is LVI-SAM also by Shan et al. [83] also known as Lidar-Visual-Inertial Odometry via Smoothing and Mapping. It combines the strengths of LIO-SAM and visual SLAM systems by integrating LiDAR, IMU and visual data for more robust localisation and mapping. This hybrid system is well-suited for lunar applications, where both dense mapping and precise localization are essential. LVI-SAM achieves comparable or superior accuracy to LIO-SAM, with success rates exceeding 95% in multi-modal terrestrial tests. Its primary limitation is computational overhead, which could challenge real-time performance in resource-constrained settings. [83]

Lastly, a robust visual-inertial odometry (VIO) framework, known as VINS-Fusion by Qin et al. [84]–[87]. It is designed to provide highly accurate state estimation by tightly fusing visual and inertial sensor data and is highly reliable in feature-sparse environments. Thus, it makes a prime candidate for navigation tasks where accurate pose estimation is critical. However, since it is just an odometry framework, provides only sparse maps natively, it requires external tools for extracting dense maps. This can potentially increase the complexity and therefore, also the computational resources required. Its success rates in localization tasks often exceed 95% in structured terrestrial environments but may decrease slightly in highly unstructured terrains without supplementary sensors. [84]–[87]

4.1.2.1 Conclusion

In summary, RTAB-Map excels in sensor flexibility and dense mapping but struggles with precision in feature-sparse areas. VINS-Fusion and LIO-SAM provide higher localization accuracy, with the former suited for visual-inertial setups and the latter better for LiDAR-based systems. ORB-SLAM3 is limited in multi-modal integration but offers strong visual localization in structured settings. LVI-SAM, as a hybrid solution, combines dense mapping and high accuracy, outperforming others in unstructured lunar environments but at the cost of computational efficiency. Due to ease of implementation, easy sensor integration/fusion and a dense mapping feature, RTAB-Map seems like the better option and is therefore used for the remainder of the project.

4.2 Tool design

Based on the research conducted on tool design and functionality in subsection 2.3.3 a variety of tools could be utilised in lunar regolith excavation. Those can be categorised into three categories: Blades

- Blades: designed for pushing loose top layer of the regolith and top layer excavation. Example S-blade
- Excavators: Scoops, buckets and rippers designed for removing deeper layers of regolith, trenching or moving loose material to another place. Example RASSOR
- Compactors: designed for compacting the regolith. Example ASPECT pins

The design requirements for such tool would be versatility of tasks it can perform like excavation, compaction and clean up, material gathering, habitat building. Additionally, weight of the tool should be minimised as much as possible has to be transported to the Moon.

Given this set of considerations and tasks that rover has to perform, which is landing site preparation. There is no single tool that can perform all tasks by itself. It is therefore necessary to use at least two or more tools. Based on table 2.7 three tools should be incorporated into the design. LANCE blade, ripper and ASPECT vibratory pins. The LANCE blade and ripper would be used to perform tasks of excavation and flattening. While ASPECT vibratory pins would be used for compaction of the regolith. Although, excavator type of tool might be deemed necessary in future tasks, the tools could be designed as modules that could be later replaced.

In consideration of the limitations in Chapter 3.1. The focus of the conceptual design will only be on versatility of the tool. Furthermore, simplicity of the design will also be prioritised and ease of mounting on the rover.

Taking that into account blade type tool should be used as design template and compactor. It's simple construction does not require complex tools to construct. Moreover, it could be designed as full passive tool which can be manually lowered to perform a task for proof of concept purposes. This is not feasible with excavators or compactors, which require an external source of power to control and operate.

5 Solution design

The following chapter transitions from the conceptual framework and design developed in the previous chapter to the implementation of this project. While the previous chapter presents what the authors consider an optimal design solution for the proposed problem, the constraints inherent to this project, such as its scope, time frame, and available resources, influence the practical implementation. Therefore, as described in section 3.1, the project is constrained to the resources available. The system for this project is designed for ROS2.

5.1 The RobuROC 4 platform

The RobuROC 4 by Robosoft (discontinued) is a four wheeled mobile robotic platform designed for military and research applications. The original platform could come with a variety of different hardware, including an robotic manipulator and other equipment. The RobuROC 4 platform utilised in this project, depicted in Figure 5.1, is outfitted with a number of pre- and post-fitted hardware. It is equipped with two major interfaces, a custom PCB and an Intel NUC NUC10i7FNH (Intel Corporation, Santa Clara, California , U.S.), with individual applications. The PCB mainly handles safety through relays, bumpers and the safety remote. The NUC is for interfacing with external sensors, drivers and general control of the system. [88]



Figure 5.1: A depiction of the Robosoft RobuROC 4 platform

The RobuROC 4 is equipped with four Mavilor BLS-072 AC Motors (Mavilor Motors, Carrer Empordà, Santa Perpètua de Mogoda, Barcelona, Spain) [89] and four ADVANCED Motion Controls DPCANTR 080B040 (ADVANCED Motion Controls, Calle Tecate, Camarillo, California, USA) (to control them). A 52.3V LiFePO4 PaqPOWER battery, fitted with a Victron smart battery sense for external monitoring, powers the platform and components through a PH300S48-24 DC-DC Converter. Safety is handled by a UD Radio Remote Control Transmitter and the PCB, through the control of a set of relays which further require interaction through the safety remote. The safety of the platform is based upon a set of bumpers. These components are depicted in Figure 5.2

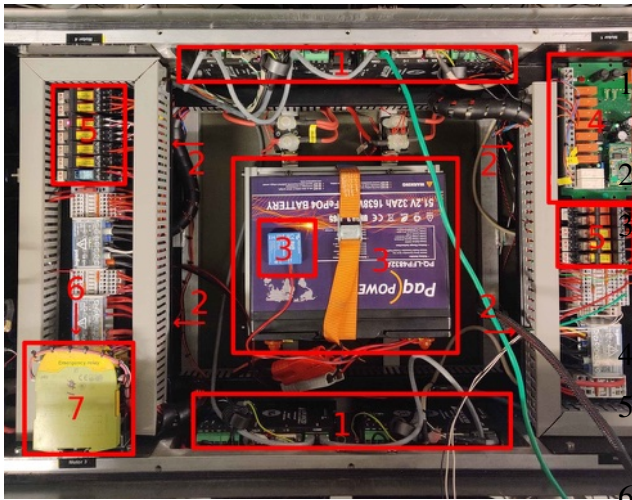
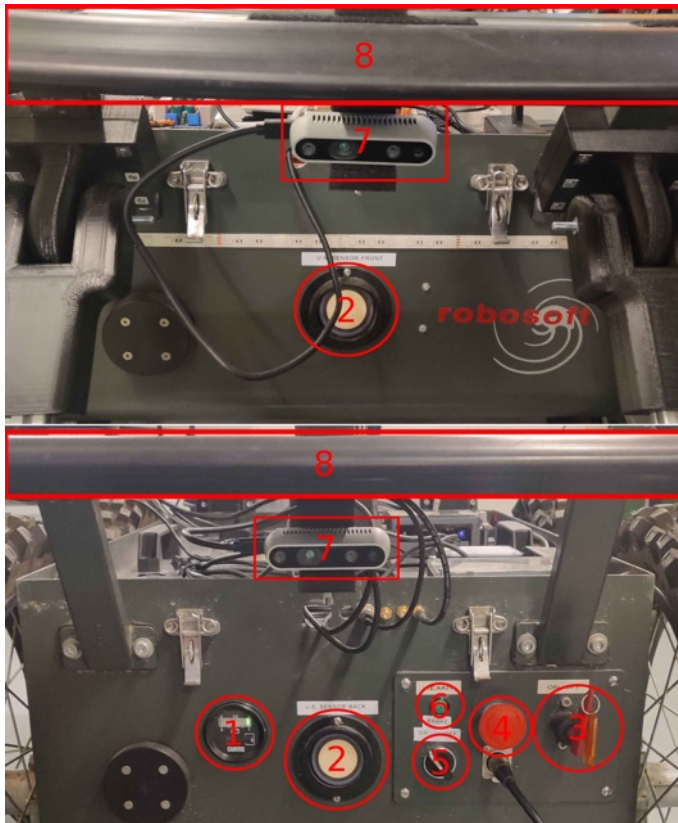


Figure 5.2: RobuROC 4 Platform internal electronics and mechanics, each component listed as its corresponding number in the list.

- 4× ADVANCED Motion Controls DPCANTR 080B040
- 4× Mavilor BLS-072 AC Motors
- 1× 48V (51,2) 28Ah 1434Wh LiFePO4 PaqPOWER battery + Victron smart battery sense [90], [91]
- 1× Custom PCB
- Multiple Interface relay RT3S4LC4 4-1415073-1 [92]
- 1× PH300S48-24 DC DC Converter [93]
- 1× PNOZ s3 24VDC 2 n/o safety relay [94]

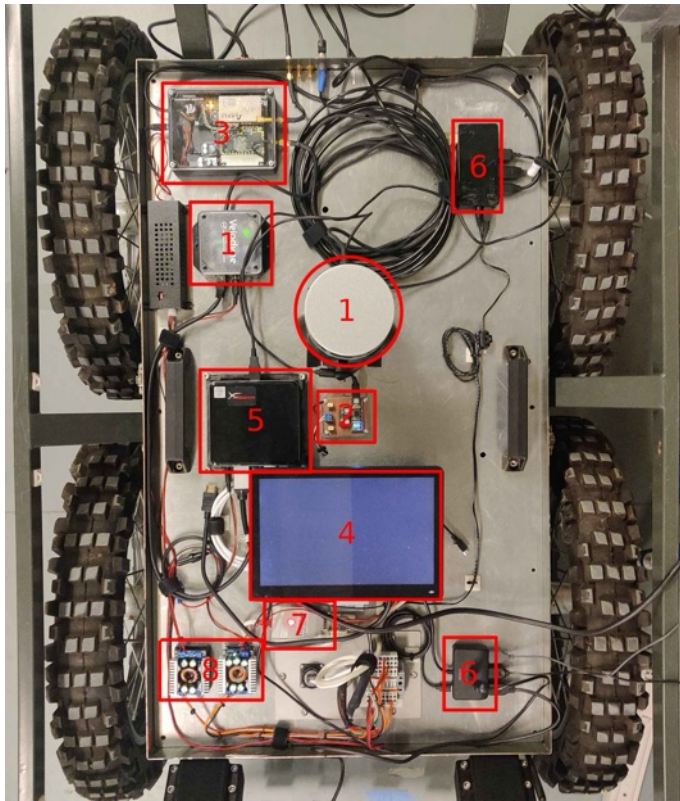
On both the front and rear of the RobuROC4, a centrally mounted ultrasonic sensor is integrated alongside an LED light strip for enhanced visibility, with the front featuring a white light strip and the rear a red one. Additionally, an air intake system is incorporated into the design. Located on the rear of the platform, the battery isolator (ignition) is positioned alongside essential control components, including an emergency stop button, a manual brake switch, and a rearming button. These components are depicted in Figure 5.3:



1. 1× 803 Battery Hour Meter 803RB2448BCJ3010 Curtis 24V/48V Circular Instrument (Similar to [95])
2. 2× UltraSonic sensors (Unknown brand)
3. Ignition (Battery isolator switch, similar to [96])
4. Emergency stop
5. Manual brake switch
6. Rearming button
7. 2× Realsense D435i RGB-D Cameras [97]
8. 2× Bumper Similar to [98]

Figure 5.3: RobuROC 4 Platform front (Upper), Back (Bottom), each component listed as its corresponding number in the list.

For sensing the RobuROC 4 is equipped with both standard and post-fitted sensors. The standard sensor elements are the Ultrasonic sensors for this particular outfitted model. Post-fitted sensors include an IMU, Magnetometer and GPS. For the purpose of this project, it is further outfitted with two Realsense D435i RGB-D Cameras on the front and back, and a Velodyne VLP-16 High-res LiDAR. The cameras are placed at a -16° angle from level, determined heuristically. The system is controlled through the Intel NUC NC10i7FNH and the PCAN interface, to which a on-board screen can be utilised for UI with the computing unit. These are all attached to a plate in the RobuROC frame, the setup can be seen depicted in Figure 5.4.



1. 1× Velodyne VLP-16 High-res [99]
2. 1× IMU HW-612 + CentIoT Nano V3 AtmegaP CH340 Micro USB [100], [101]
3. 1× Custom built RTK GPS system
4. 1× Beetrronics 10HD7 10" Screen [102]
5. 2× Intel NUC NUC10i7FNH 32 GB RAM [103]
6. 2× USB Hub (Powered)
7. 1× PCAN Interface [104]
8. 2× XL4016E1 DC-DC Step Down Buck Converter [105]

Figure 5.4: RobuROC 4 platform from above without protective lid. Each component listed as its corresponding number in the list.

Similarly, on the side of the platform, a rod is attached, which can be lowered. On this rod the magnetometer and the receiver for the RTK gps is attached.



1. 1× Magnetometer (Unknown model)

Figure 5.5: RobuROC 4 Plat form attached rod, with the Magnetometer indicated by red square

The RobuROC4 is accompanied by a DUALSHOCK 4 controller and a safety remote. The DUALSHOCK 4 controller, when paired with appropriate software, facilitates direct and intuitive control of the platform. Meanwhile, the safety remote serves critical functions, including enabling the platform for movement and controlling the activation of the LED lighting system. These and the use of the UD Radio Remote Control Transmitter can be seen in Figure 5.6.



1. 1× DUALSHOCK 4-controller
2. 1× UD Radio Remote Control Transmitter [106]

Figure 5.6: Image of the security remote (Bottom) and Dualshock 4 controller (Top).

Due to the placement and size of the wheels on the RobuROC 4, the platform is skid-steered, which will have to be considered in the final solution. Furthermore, in order to accommodate the requirement of the ability to move regolith, a tool is required to be designed for this purpose as explained in section 4.2. The choice of placement for the Realsense D435i is explained in depth in the following section.

5.1.1 Camera placement

Since the cameras are to be utilised in both mapping and localisation, the placement is essential to obtain the desired functionality. Thus, due to the general structure of the RobuROC 4, two main placements were considered, a configuration where the two cameras are placed in the front, and another where the cameras are placed on opposite ends of the platform.

Both these configurations provide benefits in regard to project and density of the map. The configu-

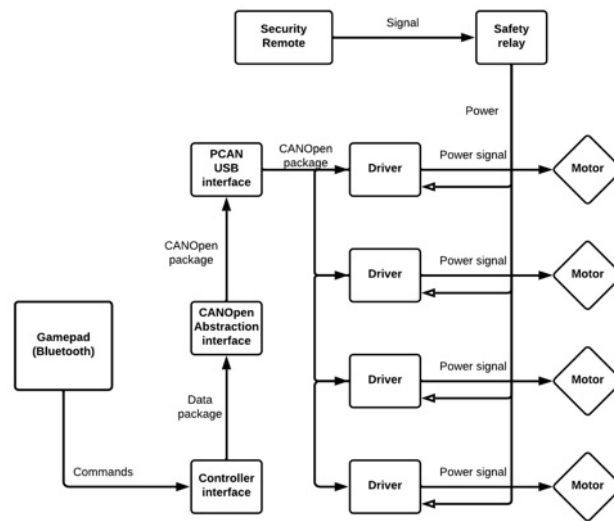


Figure 5.7: Illustration of the interactions internally for the interface between ROS2 and the RobuROC 4.

ration with both cameras in the front, provides accuracy and a wider frontal field of view. These are advantageous in context of a mobile robot only moving forward and only concerned with obstacles in front of it. Thus, regardless of the higher accuracy and widened field of view, it provides the problem that objects are only perceived from one perspective.

The other configuration provides both a frontal and back field of view, thus being able to perceive object both while traversing towards them and when moving away from them, thus providing multiple perspectives on objects within the environment. Consequently, this generates a more complete map in less time, which in this project is in high regard. Thus, the second configuration is favoured and is utilised in the remainder of this project.

5.2 System overview

With the provided platform and the additional sensor modalities added, the system needs to be designed to be compatible with these and provide the desired functionality. This and the following sections provide an overview of the proposed solution and individual components herein. The system is designed with the requirements and limitation in mind.

The system contains two major components, the RobuROC 4 interface and the SLAM module, the interacts of the first are illustrated in Figure 5.7.

In the RobuROC 4 interface, the CANOpen interface manages all data exchange through the CANOpen Peak interface, thus also the motors. The controller module is designed to manage data exchanges between the CANOpen interface and the Joy interface and potential other interfaces. It will process the data to be compatible and transfer it to the CANOpen interface.

As for the SLAM module, it consists of two methods, one using only the LiDAR for both odometry and mapping, the other using only inputs from the cameras.

For compatibility and ease of implementation the system is implemented in ROS2 Humble. The following subsection describe each of these and the interactions they have with related parts.

5.2.1 Interface

To ensure effective system operation, a comprehensive interfacing strategy is required to manage data exchange between the core platform and the individual components. The RobuROC 4 platform is equipped with a PCAN-USB interface (PEAK-System Technik GmbH, Darmstadt, Germany), which facilitates communication with the motor drivers. This interface operates using the CANopen protocol, a standardized communication framework (CiA 301) [107] designed for real-time data exchange in embedded systems. As part of this project, a dedicated communication layer will be developed to ensure seamless integration and efficient operation of the motor drivers.

In addition to the motor drivers, each sensor modality incorporated into the system requires its own interface to enable data acquisition and delivery to other subsystems. The system integrates sensors such as LiDAR, cameras, and inertial measurement units (IMUs), each of which communicates via its specific protocol. To manage this complexity, the interfacing strategy is designed to abstract hardware-specific details, providing a unified data pipeline for higher-level system components.

Key considerations for the interface design include:

- **Data access:** Ensuring ease of use and accessibility to data received and transmitted to and from sensor and other parts of the system.
- **Control:** Ensure easy control over drivers and other hardware to ensure safety of operation
- **Fail safe:** Ensure if an error occurs that the system is halted and every motorised component is stopped.

5.2.2 SLAM

To fulfill the requirements for autonomous navigation and environmental mapping, the RTAB-Map framework has been selected as the SLAM solution. RTAB-Map is particularly suited for this project due to its robust handling of large-scale environments and compatibility with multi-modal sensor inputs.

The implementation of RTAB-Map will be developed within the ROS 2 framework, ensuring modularity, scalability, and compatibility with the broader system architecture. ROS 2 is chosen for its improved real-time performance, enhanced support for multi-robot systems which are critical for the system's overall functionality. Furthermore, the ROS2 framework already incorporates RTAB-Map.

The SLAM module will integrate sensor data from the interfaces, including LiDAR and cameras, through RTAB-Map's sensor fusion capabilities. This approach enables precise localization and high-fidelity mapping by leveraging both metric and appearance-based cues. The resulting map will serve as a foundational element for navigation, obstacle avoidance, and task execution.

5.3 Tool Design

Due to delimitations for this project the main focus for tool design was targeted to excavation tool. The initial concept for the tool shown in figure 5.8, which would be mounted in front of the rover. In order to incorporate versatility into the design one edge was meant to imitate S-blade of bulldozer and rake pattern was put on the opposite edge. Moreover, to increase scooping capacity, the tool had shape of 50deg arc. Rake pattern was used incorporated to rip the ground for easier future manipulation.

Moreover the tool could be mounted at the back of the rover and used as rake.

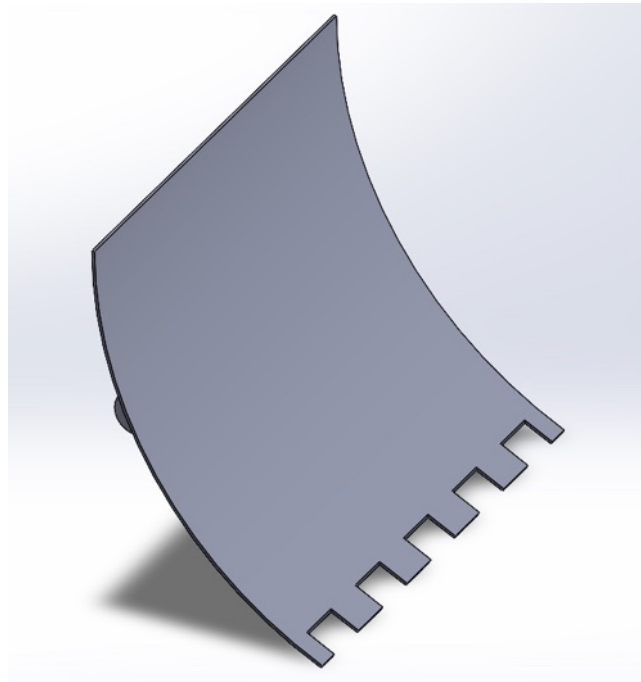


Figure 5.8: First version of tool

The initial design aimed to incorporate 5mm steel plate. Moreover to adjust the tool angle plate hinges and M10 bolts would be used to fix it to the beam and clamps would fix the whole tool securely to frame. To adjust angle of the tool hinge would be.

6 Implementation

Based on the system design, the individual components are here described in detail, including their implementation and intended functionality. The entirety of the system is implemented in ROS2 for comparability and scalability of the system. The files are mainly written in python for ease of development and readability.

The link for the github repository for this project can be found in Appendix A.

6.1 Interface

Since the proposed solution is to be implemented on the RobuROC 4, an interface to this system and the integrated sensors is required. The interfaces vary depending on the hardware used, and these are described in detail in the following sections.

6.1.1 Driver interface

A module, `roburoc_canopen`, was developed to enable efficient communication between the RobuROC4's embedded motor drives and the Robot Operating System 2 (ROS2) framework. The node abstracts low-level CANopen communication, simplifying integration with higher-level robotic control processes. The core class, `RobuROC_Canopen`, implements ROS2 topics, services, and publishers to interact with the CAN bus network. It is designed for scalability and ease of use, providing robust connection management, periodic tasks, and message subscription functionalities. The key features include:

- Simplified connection and disconnection to a CANOpen Network, adapted specifically for the RobuROC 4
- Management of periodic tasks such as heartbeat messages.
- Support for Service Data Objects (SDO) and Network Management (NMT) commands.
- Subscriptions to monitor real-time data changes from the motor drives.

The module, allows for connection to a generic physical interface or bus through the `Connect` method, a part of which is showcased in snippet 6.1.1. This searches for at least four nodes within the network, if these are not found it will fail. This is to ensure that it finds all four drivers before beginning operation, thus acting as a initial safety measure.

```
def Connect(self, bustype='pcan', channel='PCAN_USBBUS1', bitrate=1000000):
    try:
        self.CAN_NETWORK.connect(bustype=bustype, channel=channel, bitrate=bitrate)
        while len(self.CAN_NETWORK.scanner.nodes) < 4:
            self.CAN_NETWORK.scanner.search()
            sleep(1)
        for node in self.CAN_NETWORK.scanner.nodes:
            self.CAN_NODES.append(self.CAN_NETWORK.add_node(node, self._DRIVE_CONFIG))
    except Exception as e:
        self.logger.error(f"Connection failed: {e}")
```

Snippet 6.1.1: Part of the Connect method, handling scanning and integrating nodes on the network.

Mainly, this method relies on parameters that can be set in the ROS 2 framework, should the interface change. Furthermore, each node is provided the Electronic Data Sheet which acts as a library of information about the Object dictionary of the drivers and allows it to act within the bounds of this. Further methods include methods for writing, subscribing and periodic task management, prototypes of which can be seen in snippet 6.1.2.

```
def PeriodicTask(self, cobid: int, data: list, period: float = 0.2)
def RemovePeriodicTask(self, key: int)
def Subscribe(self, COBID: int, callback: callable)
def Unsubscribe(self, COBID: int, callback: callable = None)
def SDOwrite(self, node_id: int, indices: list, data: list)
def PDOwrite(self, COBID: int, data: list)
def NMTwrite(self, ctrlWord: list)
def SDOread(self, node_id: int, indices: list)
```

Snippet 6.1.2: Prototypes of methods implemented in the module for easy interfacing with the drivers

Connection, periodic tasks, object subscription and read are based on services, since they will only be utilised sporadically. However, writing will be utilised extensively and is therefore based on a regular subscriber/publisher setup. The topics and serviced associated with this module are defined as:

- **Write subscriber:** '/roburoc/CANWrite'
- **Subscription publisher:** '/roburoc/CANSubscription'
- **Connection service:** '/roburoc/CANConnection'
- **Periodic task service:** '/roburoc/CANPeriodic'
- **Subscription service:** '/roburoc/CANSubscribe'
- **Read service:** '/roburoc/CANRead'

Each of these have individual custom interfaces (ROS messages), to reduce data redundancy and data transfer overhead. The topics and services can be remapped if different names are desired. The

subscriptions created by this node require a callback to extract the data from the message that is received when the value changes, thus a generic callback is created, as seen in Snippet 6.1.3. This method just publishes all received information to the same topic, and leaves it for the user to access and interpret it by the COBID.

```
def Generic_callback(self, COBID:int, data:bytearray, rec_percentage:float):
    Subscription_message = CANSubscription()
    Subscription_message.cobid = COBID
    Subscription_message.node_id = [node for node in self.CAN_NODES if COBID >> 7 ==
    ↪ node.id][0]
    Subscription_message.data = list(data)
    self.SubscriptionPub.publish(Subscription_message)
```

Snippet 6.1.3: Generic callback utilised in the CANOpen module for all subscriptions

As this module only creates an easier interface layer, another module is employed to operate the system. Thus the module `roburoc_controller` was developed to control this module, initialise the necessary variables and thereby control the mobile platform. The control module, regardless of the interface setup, also abstracts the read, write, subscription and periodic task interfacing, see Snippet 6.1.4, although only on a ROS2 basis.

```
AddPeriodicTask(self, COBID: int, data: list, period: float)
def RemovePeriodicTask(self, COBID:int)
def Subscribe(self, COBID:int)
def Unsubscribe(self, COBID:int)
def NMT_Write(self, data: list)
SDO_Write(self, node_id: int, indices: list, data: list)
def PDO_Write(self, COBID: int, node_id: int, data: list)
def SDORead(self, node_id: int, indices: list)
```

Snippet 6.1.4: Prototypes of methods implemented in the controller module

The controller mainly concerned with the method `setSpeed`, which acts as a callback from both the gamepad and input from a navigation stack, as seen in Snippet 6.1.5. This calls one of two methods, depending on whether the message comes from the gamepad or the navigation stack.

```
def setSpeed(self, message):
    try:
        getattr(message, 'linear', self.gamepad_control(message))
    except AttributeError:
        getattr(message, 'buttons', self.navigation_control(message))
```

Snippet 6.1.5: `setSpeed` method implemented to handle two different messages

The method utilises two topics, specifically `/joy` and `/roburoc/cmd_vel`, to obtain information from either the gamepad or the navigation stack. The method further ensures that if no input is provided, that the platform is halted. This functionality is encompassed by two methods; `gamepad_control` and `navigation_control`. The first provides control through a DUALSHOCK controller, through which `'△'` acts as a deadman switch for controlling the speed of the wheels. `'□'` provides the reset functionality through the `reset` method, allowing the platform security measures to be reset. Lastly `'○'` provides a braking mechanism, that will quickly lower the speed of the platform.

The `navigation_control` provides similar functionality to the gamepad speed control functionality.

6.1.2 IMU interface

In order to interface with the Inertial Measurement Unit (IMU) and interface module is implemented. The purpose of this module is to obtain data and publish it, such that it is available to other modules within the system. The implementation of this is carried out with the use of the `serial` library, allowing access to data-streams through e.g. USB ports.

The module, amply named IMU `imu_publisher_node` utilises two ROS parameters to allow changing ports and baudrates if deemed necessary, as can be seen in Snippet 6.1.6. However, for this purpose it utilises the default port and baudrate.

```
super().__init__('imu_publisher_node')
self.declare_parameter('serial_port', '/dev/ttyUSB0') # Set your default serial
↪ port here
self.declare_parameter('baud_rate', 115200)

serial_port = self.get_parameter('serial_port').get_parameter_value().string_value
baud_rate = self.get_parameter('baud_rate').get_parameter_value().integer_value
```

Snippet 6.1.6: A snippet of the parameter setup and utilisation of the IMU module

This module is timer-based, only receiving data and publishing it at 10 Hz to the `imu/data` topic. It is composed into a `sensor_msgs.IMU` message for easy readability.

6.2 Simulation

In order to get a head-start on the implementation of sensors and other packages that would be used by system, the simulation of the RobuROC has been created using ROS2 and Gazebo. Moreover, the whole simulation would be displayed in Rviz, as seen in Figure 6.1.

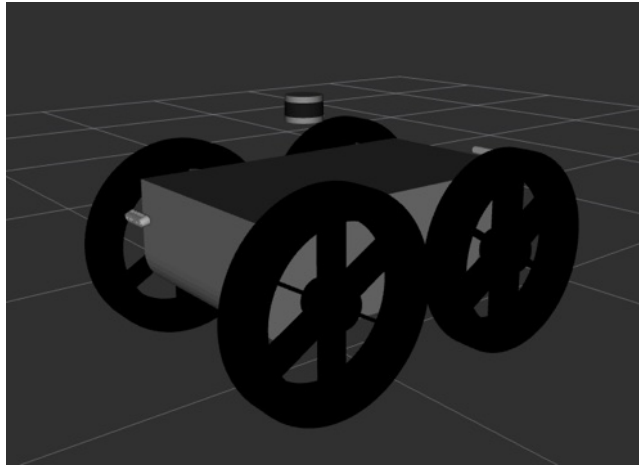


Figure 6.1: Model of the RobuROC visualised in Rviz2

6.2.1 XACRO and Gazebo model

Firstly the XACRO file was made to model the rover. For easier navigation in files and overall modelling, a master file was created that included all files that were used in the construction of the robot. As seen in code snippet 6.2.1 robot was made of five files.

```
<?xml version="1.0"?>
<robot xmlns:xacro="http://www.ros.org/wiki/xacro" name="RobuROC">
  <!-- Add all parameters -->
  <xacro:include filename="include_RobuROC.xacro" />
  <!-- Add platform -->
  <xacro:include filename="RobuROC_core.urdf.xacro" />
  <!-- Add two Realsense D435 RGB-D cameras -->
  <xacro:include filename="camera_depth.xacro"/>
  <!-- Add Lidar velodyne-16 -->
  <xacro:include filename="lidar_vlp_16.xacro" />
  <!-- Include gazebo model -->
  <xacro:include filename="RobuROC_gazebo_sim.xacro" />
</robot>
```

Snippet 6.2.1: Master file of the RobuROC

include_RobuROC.xacro

The main purpose of this file is to contain all variables of the robot in one file. This include physical properties of the robot like the dimensions and mass of rovers parts, translation and rotation of frames. This parametrised approach made whole model more flexible and making easier to making any adjustments. Moreover, equations for calculations of moment of inertia for simple shapes has been made. In snippet 6.2.2 the properties for each part can be seen.

```

<!-- ##### -->
<!--          Chassis          -->
<!-- ##### -->
<xacro:property name="box_y" value="0.54" />
<xacro:property name="box_x" value="0.925" />
<xacro:property name="box_z" value="0.304" />
<xacro:property name="box_m" value="100" />
<xacro:property name="box_off_z" value="0.484" />
<xacro:property name="chassis_vis_origin" value="0 0 ${box_off_z}" />
<xacro:property name="chassis_col_origin" value="0.0 0 ${box_off_z-box_z/2}" />
<!-- ##### -->
<!--          Wheels          -->
<!-- ##### -->
<xacro:property name="cyl_l" value="0.1" />
<xacro:property name="cyl_r" value="0.28" />
<xacro:property name="cyl_m" value="10" />
<xacro:property name="cyl_off_x" value="0.3068" />
<xacro:property name="cyl_off_y" value="0.3425" />
<xacro:property name="cyl_off_z" value="${box_off_z-0.204}" />
<xacro:property name="rot_90" value="${pi/2}" />
<xacro:property name="max_vel" value="2" />
<xacro:property name="wheel_1_origin" value="${cyl_off_x} ${cyl_off_y} ${cyl_off_z}"/>
<xacro:property name="wheel_2_origin" value="${cyl_off_x} ${-cyl_off_y} ${cyl_off_z}"/>
<xacro:property name="wheel_3_origin" value="${-cyl_off_x} ${-cyl_off_y} ${cyl_off_z}"/>
<xacro:property name="wheel_4_origin" value="${-cyl_off_x} ${cyl_off_y} ${cyl_off_z}"/>
<!-- ##### -->
<!--          STL          -->
<!-- ##### -->
<xacro:property name="scale_factor" value="0.001" />
<xacro:property name="scale" value="${scale_factor} ${scale_factor} ${scale_factor}"/>
<!-- ##### -->
<!--          Camera          -->
<!-- ##### -->
<xacro:property name="camera_off_x" value="0.6" />
<xacro:property name="camera_off_z" value="0.45" />
<xacro:property name="camera_rot" value="${16*pi/180}" />
<xacro:property name="camera_1_origin" value="${camera_off_x} 0 ${camera_off_z}" />
<xacro:property name="camera_2_origin" value="${-camera_off_x} 0 ${camera_off_z}" />
<xacro:property name="camera_1_rotation" value="0 ${camera_rot} 0" />
<xacro:property name="camera_2_rotation" value="0 ${camera_rot} ${pi}" />
<!-- ##### -->
<!--          Lidar          -->
<!-- ##### -->
<xacro:property name="lidar_off_x" value="0.12" />
<xacro:property name="lidar_off_z" value="${box_off_z+0.155}" />
<xacro:property name="lidar_origin" value="${lidar_off_x} 0 ${lidar_off_z}" />

```

RobuROC_core.urdf.xacro

This file creates whole base frame of the rover without any sensors. The collision was modelled by using simple shapes box for chassis and cylinders for wheels. Additionally, the chassis and wheels were modelled in SOLIDWORKS saved in STL format and used for visual purposes. As the meshes were saved in millimeter format they had to be scale down by factor of 0.001, Snippet 6.2.3 shows an example of how the chassis was modelled. The chassis joint was placed in point (0,0,0) as it would be easier for future navigation purposes that robot's initial frame start at this point.

```

<!-- ##### -->
<!--          Chassis Link          -->
<!-- ##### -->

<!-- Chassis joint has been off seted in z axis and set to fixed-->
<joint name="chassis_joint" type="fixed">
  <parent link="base_link"/>
  <child link="chassis_link"/>
</joint>

<link name="chassis_link">
  <visual>
    <origin xyz="${chassis_vis_origin}" rpy="${rot_90} 0 ${rot_90}"/>
    <geometry>
      <mesh filename="file://$(find RobuROC_sim)/meshes/chassis.STL"
        ↪ scale="${scale}"/>
    </geometry>
    <material name="grey" />
  </visual>
  <collision>
    <origin xyz="${chassis_col_origin}" rpy="0 0 0"/>
    <geometry>
      <box size="${box_x} ${box_y} ${box_z}" />
    </geometry>
  </collision>
  <xacro:inertial_box mass="${box_m}" x="${box_x}" y="${box_y}" z="${box_z}">
    <origin xyz="${chassis_col_origin}" rpy="0 0 0"/>
  </xacro:inertial_box>
</link>

```

Snippet 6.2.3: Example of chasis joint and link

camera_depth.xacro

In order to add two realsense cameras into the model, the camera_depth.xacro file has been created. It utilises the URDF files provided by Intel in ROS2 package realsense2. As only frames of all camera lenses and visual aspect was available from realsense2 package. A third party code was used to firstly model the cameras in Gazebo simulator and then by use of gazebo plug-in ros topics mimicking data

from cameras would be send. Example of adding one camera and gazebo model can be seen in snippet 6.2.4.

```
<xacro:sensor_d435 parent="base_link" name="camera1" use_nominal_extrinsics="$(arg
↪ use_nominal_extrinsics)" add_plug="$(arg add_plug)" use_mesh="$(arg use_mesh)">
  <origin xyz="$(camera_1_origin)" rpy="$(camera_1_rotation)"/>
</xacro:sensor_d435>

<!-- Realsense Gazebo Plugin -->
<xacro:property name="name" value="camera1" />
<xacro:property name="topics_ns" value="camera1" />
<xacro:property name="publish_pointcloud" value="true"/>

<xacro:gazebo_d435 camera_name="$(name)" reference_link="$(name)_link"
↪ topics_ns="$(topics_ns)" depth_optical_frame="$(name)_depth_optical_frame"
↪ color_optical_frame="$(name)_color_optical_frame"
↪ infrared1_optical_frame="$(name)_left_ir_optical_frame"
↪ infrared2_optical_frame="$(name)_right_ir_optical_frame"
↪ publish_pointcloud="$(publish_pointcloud)">
</xacro:gazebo_d435>
```

Snippet 6.2.4: Example of the code of how front camera was added to model

lidar_vlp_16.xacro

Similarly to adding cameras the Velodyne lidar used in the project was added by using third party code. It already contained a Gazebo model with designed plug-in to mimic published by lidar topics. As the lidar used in project was high resolution with 20° angular FoV the gazebo file was changed accordingly. Snippet 6.2.5 shows implementation of the lidar to model.

```
<xacro:include filename="VLP-16.urdf.xacro" />
<xacro:VLP-16 parent="base_link" name="velodyne" topic="velodyne_points" hz="10"
↪ samples="440" gpu="$(gpu)">
  <origin xyz="$(lidar_origin)" rpy="0 0 0" />
</xacro:VLP-16>
```

Snippet 6.2.5: Example of how lidar was add to the robot model

RobuROC_gazebo_sim.xacro

For Gazebo simulator to simulate objects accurately it needs information about collision, mass, inertia and friction forces. First three were already defined in XACRO file. Therefore friction forces has to be defined for chassis and wheels. Moreover gazebo is sending back to Rviz joint state of each wheel. Additionally in order to move rover in the simulator gazebo plug-in for differential drive was utilised as seen in snippet 6.2.6. Additionally differential drive outputs odometry frame that could later be utilised by mapping algorithm.

```

<!-- ##### -->
<!-- Differential drive -->
<!-- ##### -->
<gazebo>
  <!-- Gazebo plugin for differential drive -->
  <plugin name="skid_steer_drive" filename="libgazebo_ros_diff_drive.so">
    <ros>
      <namespace> </namespace>
    </ros>

    <num_wheel_pairs>2</num_wheel_pairs>

    <!-- Wheels 0 -->
    <left_joint>wheel_1_joint</left_joint>
    <right_joint>wheel_2_joint</right_joint>

    <!-- Wheels 1 -->
    <left_joint>wheel_4_joint</left_joint>
    <right_joint>wheel_3_joint</right_joint>

    <!-- Kinematics -->
    <wheel_separation>${box_y+cyl_l}</wheel_separation>
    <wheel_separation>${box_y+cyl_l}</wheel_separation>

    <wheel_diameter>${cyl_r*2}</wheel_diameter>
    <wheel_diameter>${cyl_r*2}</wheel_diameter>

    <!-- Limits -->
    <max_wheel_torque>1000</max_wheel_torque>
    <max_wheel_acceleration>5.0</max_wheel_acceleration>

    <!-- Output -->
    <publish_odom>true</publish_odom>
    <publish_odom_tf>true</publish_odom_tf>
    <publish_wheel_tf>true</publish_wheel_tf>

    <odometry_frame>odom</odometry_frame>
    <robot_base_frame>base_link</robot_base_frame>

  </plugin>
</gazebo>

```

Snippet 6.2.6: Differential drive setup in gazebo

6.2.2 Simulation launch

Once the model was deemed satisfactory it was spawned in Gazebo simulator in prepared earlier worlds. First world was made of available objects like bookshelves, tires, dumpsters and coffee tables figure 6.2.

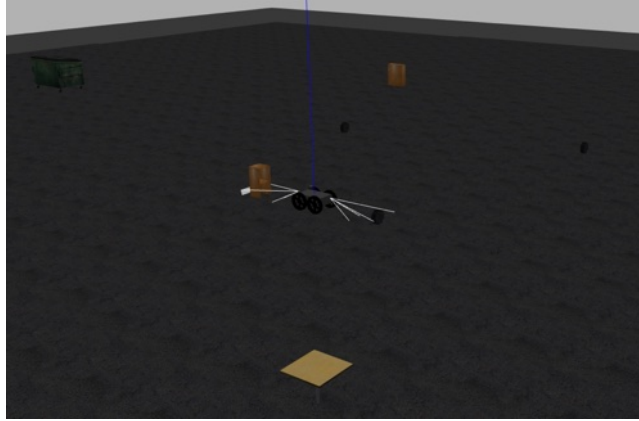


Figure 6.2: First environment created in Gazebo with spawned RobuROC

The purpose of this world was to test if Rviz is properly receiving data from sensors modelled in Gazebo. As seen in figures 6.3, 6.4

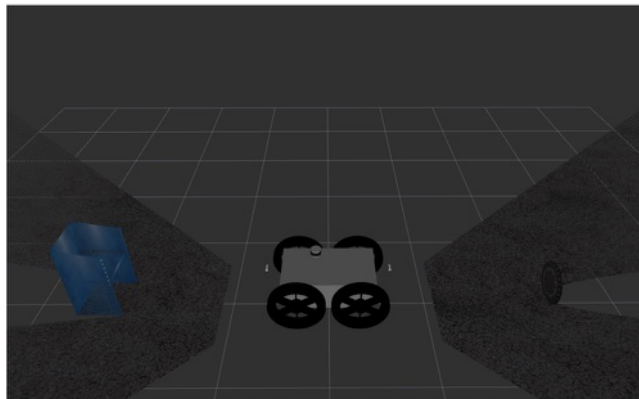


Figure 6.3: Realsense camera pointcloud visualisation in Rviz

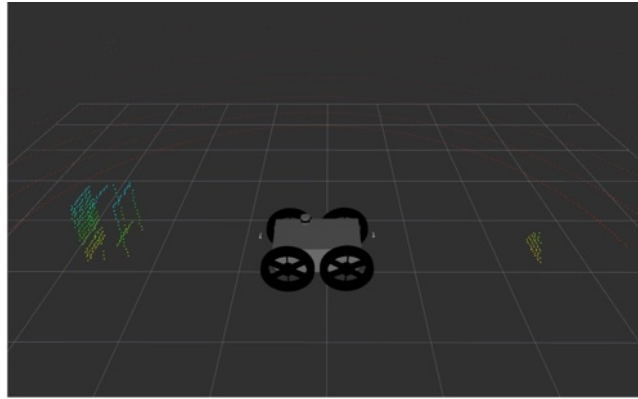


Figure 6.4: Lidar pointcloud visualisation in Rviz

For further testing of the whole system second environment in gazebo was made that utilised existing in Gazebo moon landscape figure 6.5.

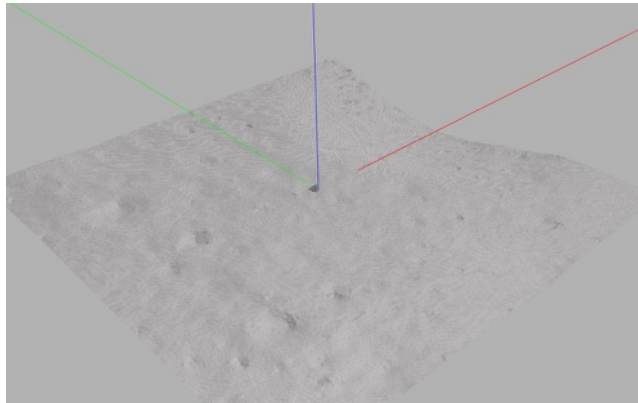


Figure 6.5: Second environment created in Gazebo without spawned RobuROC

Due to landscape not being flat the spawning node in launch file was modified to spawn robot above the ground. Due to limited time it was decided to stop simulation development and pursue implementation of the XACRO model onto the platform.

6.3 SLAM

The structure of the implemented SLAM method is organized with ROS2 nodes interacting with each other. The nodes are implemented in separate Python launchfiles and launched sequentially in a master launchfile.

6.3.1 Prerequisites

Before system will be able to map environment the sensors on board have to be launched. This subsections explains how the system was set up before RTAB-Map is used.

Camera calibration

Prior to use both RGBD cameras for SLAM, it is essential to calibrate them to enhance their performance regarding the specific lighting conditions and distance of the camera relative to the mapped area. The on-chip calibration available in the Realsense viewer was used to calibrate both cameras.

LiDAR drivers

To receive data from the Velodyne-16, a driver node is first launched to establish a connection with the LiDAR. The data that is received from the LiDAR is then converted into usable data and data types for the ROS2 communication framework through the `velodyne_pointcloud` node. The data is then published to the `velodyne_points` ros topic. Both nodes can be seen in Snippet 6.3.1

```
Driver_vel = IncludeLaunchDescription(
    PythonLaunchDescriptionSource([os.path.join(
        get_package_share_directory('velodyne_driver'), 'launch'),
        '/velodyne_driver_node-VLP16-launch.py']],
    )
Pointcloud_vel = IncludeLaunchDescription(
    PythonLaunchDescriptionSource([os.path.join(
        get_package_share_directory('velodyne_pointcloud'), 'launch'),
        '/velodyne_transform_node-VLP16-launch.py']],
    )
```

Snippet 6.3.1: The driver and pointcloud nodes in launch file

RGBD cameras node

In order to communicate with the Realsense D435i's the `rs_launch.py` launch file, provided by Intel, is utilised. this launch file interfaces with both cameras, providing them with individual IDs and providing access to various parameters.

Each camera then publishes the following topics, providing access to the output from the individual camera functions.

- `/color/camera_info`
- `/color/image_raw`
- `/depth/camera_info`
- `depth/image_rect_raw`

In order to differentiate between the two cameras, namespaces `camera1` and `camera2` respectively, are added to the topic names.

Sync nodes

RTAB-map provides synchronisation nodes, that allows the system to to merge the three needed messages coming from each camera; `color/image_raw`, `depth/image_rect_raw`, and `color/camera_info` (intrinsic) messages. This is to ensure that frames from each topic are aligned in both size and has the same timestamp.

```
# Sync node
rgbd_sync1_node = launch_ros.actions.Node(
    package='rtabmap_sync', executable='rgbd_sync', name='rgbd_sync1', output="screen",
    parameters=[{
        "approx_sync": False,
    }],
    remappings=[
        ("rgb/image", "camera1/color/image_raw"),
        ("depth/image", "camera1/depth/image_rect_raw"),
        ("rgb/camera_info", "camera1/color/camera_info"),
        ("rgbd_image", 'rgbd_image')],
    namespace='camera1'
)
```

Snippet 6.3.2: Sync node implementation for camera1. Takes as input the color/image_raw, depth/image_rect_raw and camera_info messages and outputs one fused, synchronized rgbd_image message.

The approx_sync parameter is set to False as the the subscribed topics had same source. Therefore it was unnecessary to approximate the synchronisation while grouping the data.

6.3.2 Odometry

The available exteroceptive sensors allow for RGBD and Iterative Closest Point (ICP) odometry. RTAB-Map provides the rtabmap_odom package which includes pre-implemented executables for both odometry types, namely rgbd_odometry and icp_odometry. The odometry node matches one of those executables with the right parameters (RGBD or ICP) and topic remappings. It then publishes a standardized /odom topic for both cases.

RGBD odometry

Although RTAB-Map is designed to fuse information from multiple RGB-D cameras, this functionality was not successfully implemented in this project. By default RTAP-Map uses ORB and GFTT for feature-extraction and description. For loop-closure it utilises **Tree-based newORk Optimizer** (TORO) by default. Ultimately, only a single RGB-D sensor was utilized for odometry, while both sensors contributed to mapping. This was accomplished by remapping the input topic for the RGB-D odometry node from rgbd_image to /camera1/rgbd_image

Here is a simplified version of the RGBD odometry node. RGBD odometry parameters are set to default and therefore not shown here.

```
# RGB-D odometry
rgbd_odometry_node = launch_ros.actions.Node(
    package='rtabmap_odom', executable='rgbd_odometry', output="screen",
    parameters=[{
        "frame_id": 'base_link',
        "odom_frame_id": 'odom',
        "subscribe_rgbd": True
        # (Default RGB-D odometry parameters)
    }],
    remappings=[
        ("rgbd_image", '/camera1/rgbd_image'),
        ('odom', '/odom')
    ]
)
```

Snippet 6.3.3: RGBD odometry node implementation. Takes as input the `rgbd_image` message containing the 3 RGBD topics grouped in the sync node.

In this snippet :

- **'subscribe_rgbd': True:** subscribes the odometry node to the global, synchronized `rgbd_image` message coming from sync nodes instead of the separated RGBD topics.

ICP odometry

ICP odometry, utilises the prior point cloud and the input point cloud, and attempts to minimize the distance between them iteratively to determined the new position. The data is obtained through the `velodyne_points` topic, containing the raw point cloud from the LiDAR, and is are used in this context.

Snippet 6.3.4 shows implementation of the odometry.

```
# Lidar odometry
scan_cloud = launch_ros.actions.Node(
    package='rtabmap_odom', executable='icp_odometry', output='screen',
    parameters=[{
        'frame_id': 'base_link',
        'odom_frame_id': 'odom',
        # (Default ICP parameters)
    }],
    remappings=[
        ('scan_cloud', '/velodyne_points'),
        ('odom', '/odom'),
    ]
)
```

Snippet 6.3.4: ICP odometry node implementation. Takes as input the `velodyne_points` pointcloud from the LiDAR.

SLAM

The `rtabmap_slam` is the SLAM node of the RTAB-Map package. It's main use in this project was to map the environment. The main inputs are `/odom` which is output from odometry node and sensor that has been used for mapping. Multiple combinations of odometry and sensor input were attempted, but only two configurations proved to be functional. First, RGB-D odometry using one camera and RGB-D mapping using two cameras. Second ICP odometry and LiDAR mapping.

Implementation 1 : RGB-D based SLAM

This implementation is concerned with the configuration using a single camera for odometry and two cameras for mapping. Snippet 6.3.5 shows code implementation of this approach. Key parameters for dual RGB-D mapping are as follow:

```
# SLAM
slam_node = launch_ros.actions.Node(
    package='rtabmap_slam', executable='rtabmap', output="screen",
    parameters=[{
        "rgbd_cameras": 2,
        "subscribe_rgbd": True,
        'subscribe_scan_cloud': False,
        "subscribe_odom_info": True,
        'odom_frame_id': 'odom',
        "frame_id": 'base_link',
        "map_frame_id": 'map',
        "approx_sync": True,
        "map_always_update": True,
    }],
    remappings=[
        ("rgbd_image0", '/camera1/rgbd_image'),
        ("rgbd_image1", '/camera2/rgbd_image'),
        ("odom", 'odom'),
    ],
    arguments=["--delete_db_on_start"],
    prefix='',
    namespace='rtabmap'
)
```

Snippet 6.3.5: Dual RGBD SLAM implementation.

- **'rgbd_cameras': 2** : Specifying to RTAB-Map that 2 cameras are used as input for SLAM. Note that RTAB-Map has to be built from source with the `-DRTABMAP_SYNC_MULTI_RGBD=ON` parameter to allow the use of more than one camera.
- **'subscribe_scan_cloud' : False** : Stating that only the RGBD input stream is used for SLAM.
- **'approx_sync' : True** : Allows for slight timestamp differences between odometry and mapping data messages. This was found to be necessary for the proper functioning of the SLAM algorithm.

- **'map_always_update' : True** : Always update the occupancy grid map even if the graph has not been updated (e.g., by default when the robot is not moving, the graph is not growing). Even though this parameter is initially meant to be set for idle robots that have to map dynamic environments, it was noticed that it also forces the map to be more regularly updated when moving at low speeds, resulting in denser maps of the ground surface.

Implementation 2 : LiDAR based SLAM

This implementation is concerned with the configuration only utilising the LiDAR for both odometry and mapping. Snippet 6.3.6 shows the implementation of the LiDAR based SLAM. Main difference is change of `subscribe_scan_cloud` to `True` which subscribes to pointcloud generated by LiDAR. It should be noticed that this node also subscribes to RGB-D topics, which should reinforce mapping by providing data from the cameras.

```
# SLAM
slam_node = launch_ros.actions.Node(
    package='rtabmap_slam', executable='rtabmap', output="screen",
    parameters=[{
        "rgbd_cameras": 2,

        'odom_frame_id': 'odom',
        "frame_id": 'base_link',
        "map_frame_id": 'map',

        "subscribe_rgbd": True,
        'subscribe_scan_cloud': True,
        "subscribe_odom_info": True,

        "approx_sync": True,

    }],
    remappings=[
        ("rgbd_image0", '/camera1/rgbd_image'),
        ("rgbd_image1", '/camera2/rgbd_image'),
        ("odom", 'odom'),
        ('scan_cloud', '/velodyne_points')],
    arguments=["--delete_db_on_start"],
    prefix='',
    namespace='rtabmap'
)
```

Snippet 6.3.6: LiDAR based mapping

- **'rgbd_cameras': 2** : Specifying to RTAB-Map that 2 cameras are used as input for SLAM. Note that RTAB-Map has to be built from source with the `-DRTABMAP_SYNC_MULTI_RGBD=ON` parameter to allow the use of more than one camera.

- **'subscribe_scan_cloud' : True** : Stating that only the RGBD input stream is used for SLAM.
- **'subscribe_rgb' : True** : Stating that only the RGBD input stream is used for SLAM.
- **'approx_sync' : True** : Allows for slight timestamp differences between odometry and mapping data messages. This was found to be necessary for the proper functioning of the SLAM algorithm.
- **'map_always_update' : True** : Always update the occupancy grid map even if the graph has not been updated (e.g., by default when the robot is not moving, the graph is not growing). Even though this parameter is initially meant to be set for idle robots that have to map dynamic environments, it was noticed that it also forces the map to be more regularly updated when moving at low speeds, resulting in denser maps of the ground surface.

RTAB-Map does not support loop closure without visual input [108]. Therefore, this second implementation is more subject to map drift over time since no global graph optimization is performed after registration.

6.4 Tool

Due to delimitations for this project the main focus for tool design was targeted to excavation tool. The initial concept for the tool shown in figure 6.6, which would be mounted in front of the rover. In order to incorporate versatility into the design one edge was meant to imitate S-blade of bulldozer and rake pattern was put on the opposite edge. Moreover, to increase scooping capacity, the tool had shape of 50deg arc. Rake pattern was used incorporated to rip the ground for easier future manipulation. Moreover the tool could be mounted at the back of the rover and used as rake.

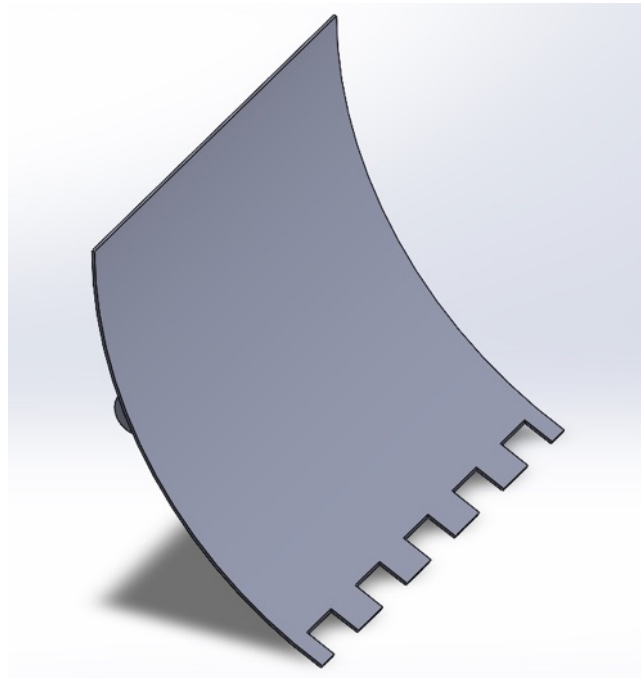


Figure 6.6: First version of tool

The initial design aimed to incorporate 5mm steel plate. Moreover to adjust the tool angle plate hinges and M10 bolts would be used to fix it to the beam and clamps would fix the whole tool securely to

frame. To adjust angle of the tool hinge would be.

Due to availability of materials in Univeristy's workshop and time it was decided to use 3mm aluminium sheet. Compared to initial design 5.3 the hinge was moved to frame fixture. This way it would be possible to adjust height of the tool while plate would be more rigidly fixed at fixed angle, The Drawing of the unbent plate can be seen in Figure 6.7.

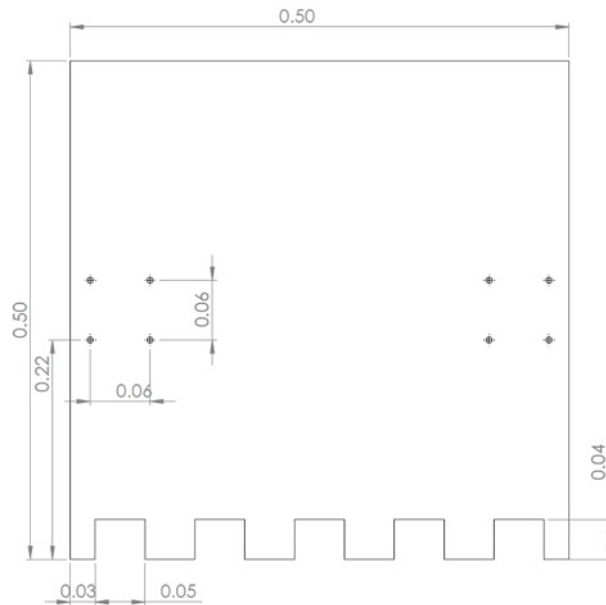


Figure 6.7: CAD model of plate fixture

Additionally fixtures for the plate in two variants 6.8, 6.9 and rover's 6.10 frame would be 3D printed in PLA. Although Nylon Carbon printer was available in Makerspace PLA printer were more available and had bigger printing plate which could fit more than one part resulted in overall shorter time of printing.

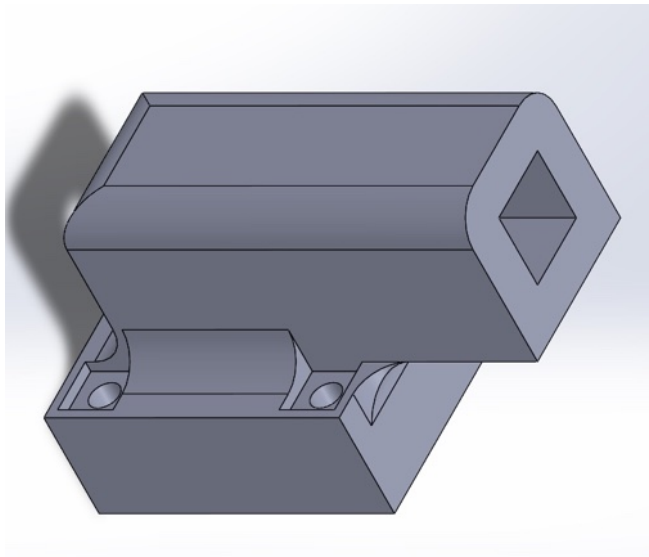


Figure 6.8: CAD model of plate fixture 50° variant

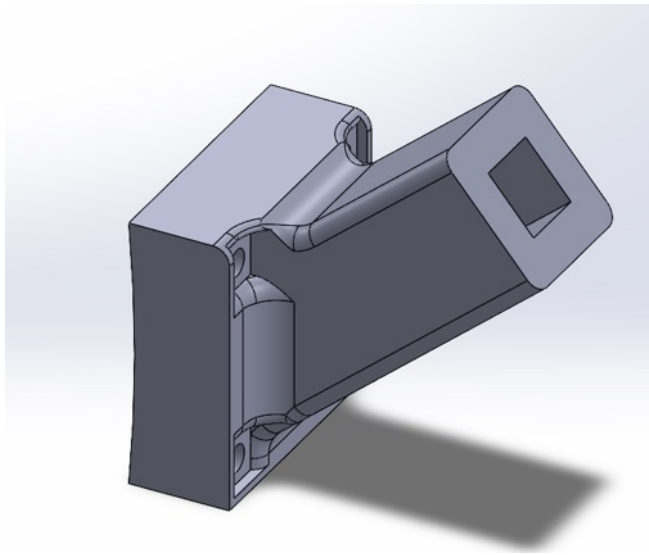


Figure 6.9: CAD model of plate fixture 90° variant

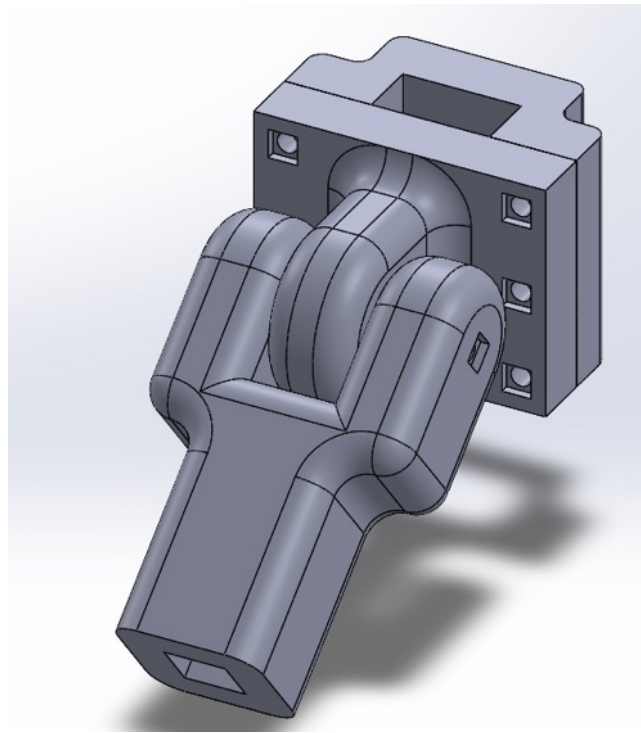


Figure 6.10: CAD assembly of frame fixture

Two variants for the fixture were made as it was unclear if the camera would be obstructed by the tool. Extruded aluminium beam 20mmx20mm was used to connect both plate and frame fixture. Final assembly of tool fixture can be seen in Figure6.11

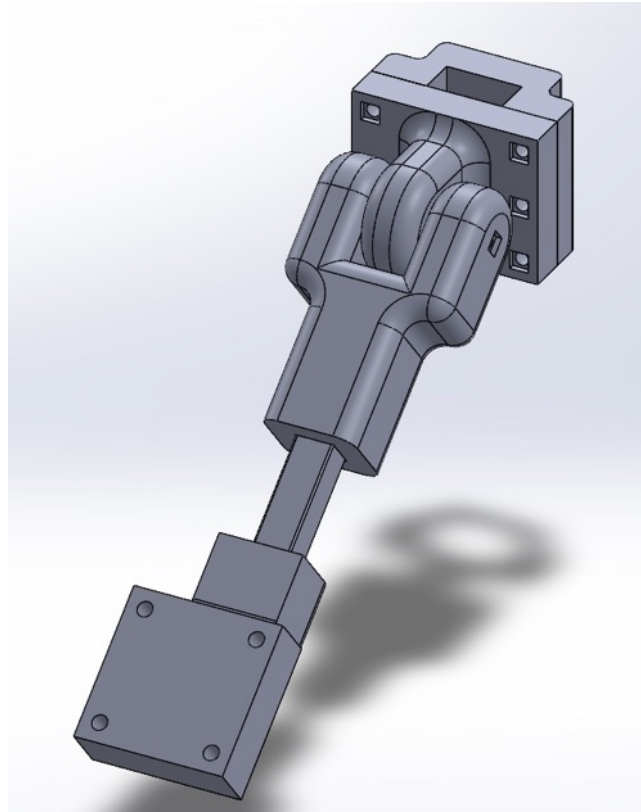


Figure 6.11: Final CAD assembly without plate

Chosen 3mm plate was too difficult to bend with available tools therefore it was decided to use flat plate. Initial fitting proved to be partially successful. All parts of fixtures fit properly and there was no issues. However as seen in Figure 6.12 the not bent sheet was obstructing the front RGB-D camera in both fixture variants.



Figure 6.12: First assembly of the tool with 50° configuration

Therefore it was decided to cut the sheet just above hole fixtures. Final assembly of the tool can be seen 6.13

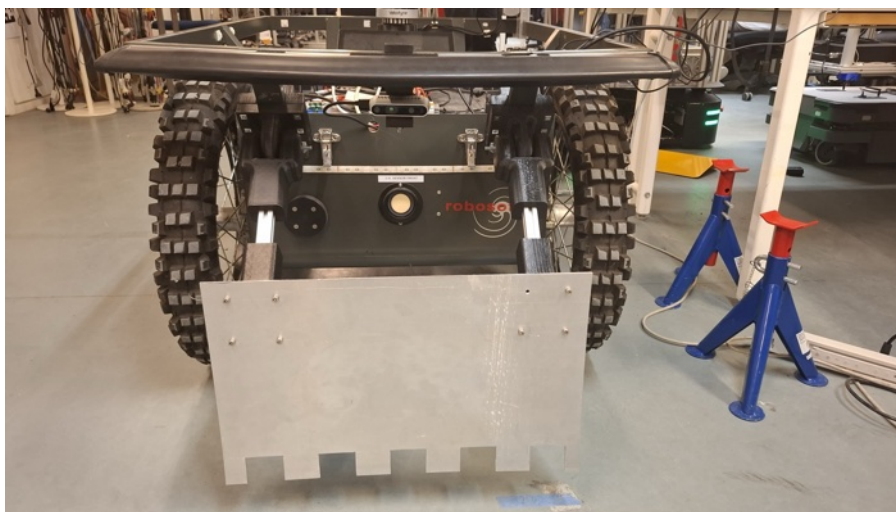


Figure 6.13: Final tool assembly with 90° configuration

7 Testing

In order to check and evaluate the rover's performance a number of tests have been designed. All tests have been designed in way to check if objectives in Chapter 3. The tests were divided into two categories SLAM and Tool. Subsequent sections describe the testing scenario and results.

7.1 SLAM algorithm

In order to test RTAB-map SLAM algorithm three scenarios with ranging difficulty were created. The aim of all tests was for the rover to map the environment and localise itself in it. The variable is scarcity of the environment. The test would be conducted in these places:

- **Robotics lab** - Environment with most features



Figure 7.1: Testing place: Robotics laboratory

- **Courtyard** - Environment with less features



Figure 7.2: Testing place: Courtyard

- **Football pitch** - Open field with little amount of features



Figure 7.3: Testing place: Football pitch

RobuROC was controlled by a DUALSHOCK 4 controller during all tests. Maximum speed of the rover was 20 cm/s and was kept constant. In addition, a GPS RTK system was used to determine the position of the rover and was used as ground truth during the tests in the courtyard and on the football pitch. Two different SLAM algorithms were also tested at each test location, described in Chapter 6.3. Each implementation of the algorithm was tested twice at each test site.

Data from ROS2 was gathered using rosbag command that subscribes and saves desired topics. Additionally the GPS data was recorded using u-center software on separate laptop that was placed on top of the rover.

7.1.1 Results

Due to a significant errors in the data collected, including missing data and calibrating the GPS module, one a single test for each odometry type is present for each environment. The tests are individually evaluated and showcased below

Laboratory

The laboratory was considered a structured and feature-rich environment for testing the system. The tests were conducted in the controlled environment of the laboratory at Aalborg University. The procedure involved driving the robot from a central position in the lab to the entrance, proceeding to the back of the lab, and subsequently returning to the central position.

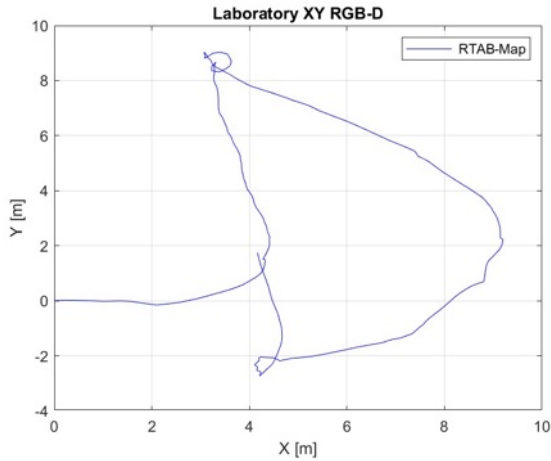


Figure 7.4: Path of the robot based on RGB-D map- ping in Laboratory

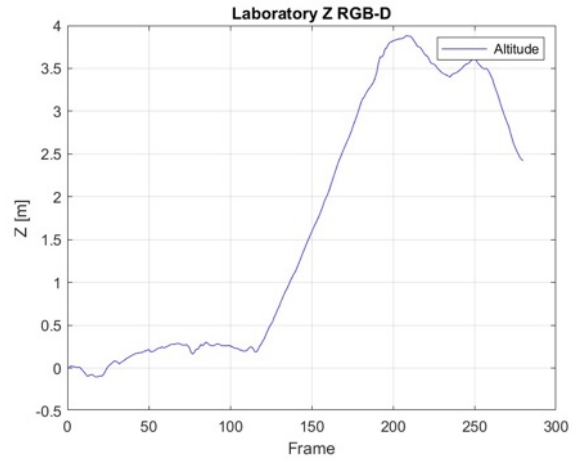


Figure 7.5: Altitude of the robot based on RGB-D mapping in Laboratory

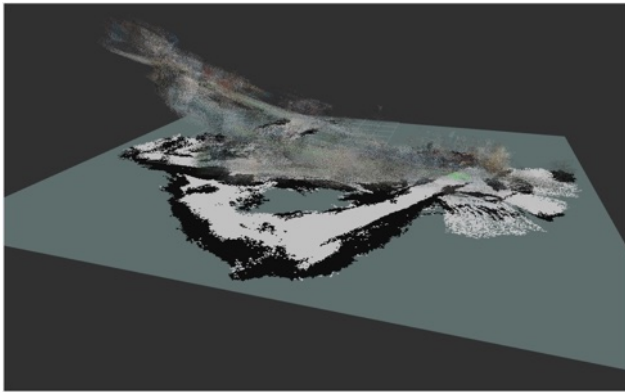


Figure 7.6: Map and pointcloud visualised in Rviz based on RGB-D mapping in Laboratory

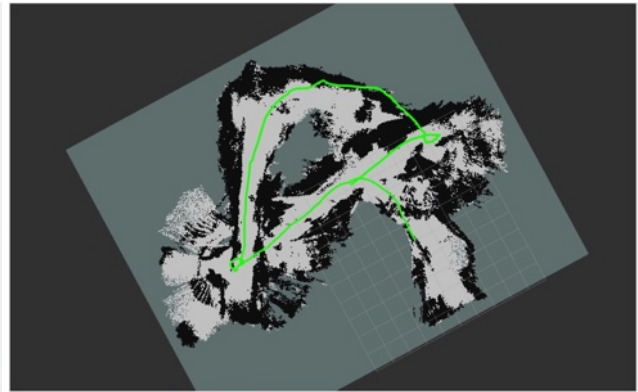


Figure 7.7: Map and path visualised in Rviz based on RGB-D mapping in Laboratory

As can be seen in Figure 7.4 and 7.5 the RGB-D odometry does not perform that well in the laboratory, which is attributed to the reflective surfaces present there. It generally loses its bearings during turns and especially in 180° turns as seen in this case. This resulting in the maps presented in Figure 7.6 and 7.7. In addition, the odometry also calculates a vertical movement, which is not consistent with what is seen. Consequently, due to the odometry losing track, the mapping process also fails

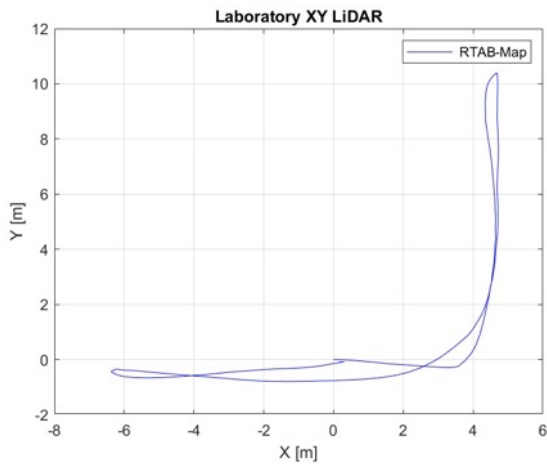


Figure 7.8: Path of the robot based on LiDAR mapping in Laboratory

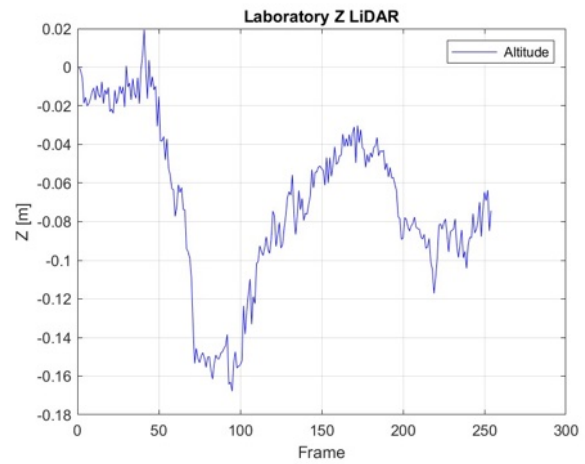


Figure 7.9: Altitude of the robot based on LiDAR mapping in Laboratory

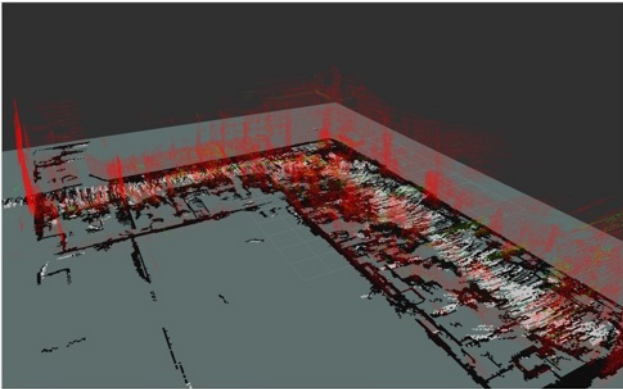


Figure 7.10: Map and pointcloud visualised in Rviz based on LiDAR mapping in Laboratory

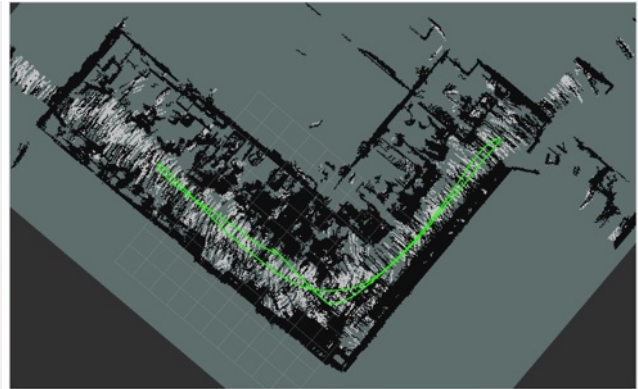


Figure 7.11: Map and path visualised in Rviz based on LiDAR mapping in Laboratory

Contrary to RGB-D, the ICP odometry method performed well in the laboratory, as can be seen in Figure 7.8 and 7.9. Although no vertical movement is perceived during testing, it keeps this error relatively low. As for the mapping, it keeps a consistent map with the correct shape. However, due to it being a dynamic environment, non-existent obstacles are scattered throughout the map, as seen in Figure 7.10 7.11

Courtyard

The courtyard tests were conducted in the courtyard of the Faculty of Technology, IT, and Design at Aalborg University. This environment offers a larger, more structured, and less feature-dense setting compared to the laboratory. The tests involved guiding the robot along the edge of half the grassy area in the courtyard, navigating around a sculpture, and returning to the original starting position.

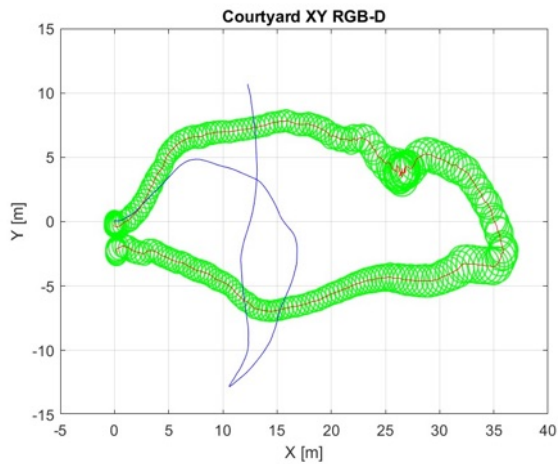


Figure 7.12: Path of the robot based on RGB-D mapping in Courtyard.

Blue line indicates RTAB-Map odometry, Red GPS signal and green circles show 2D accuracy of the GPS

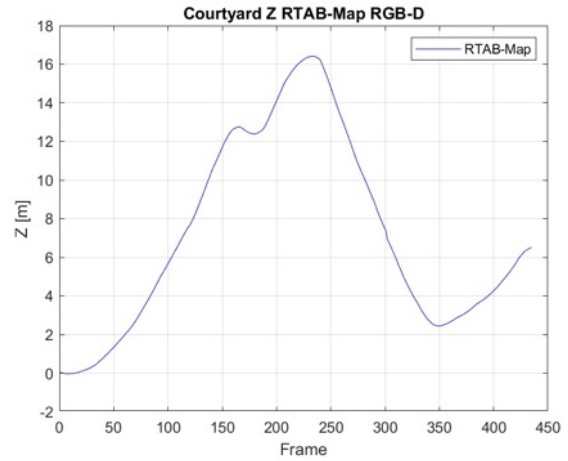


Figure 7.13: Altitude of the robot based on RGB-D

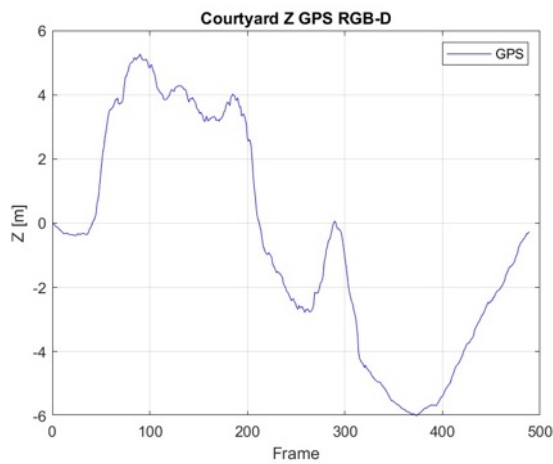


Figure 7.14: Altitude of the robot based on GPS during RGB-D mapping in Courtyard

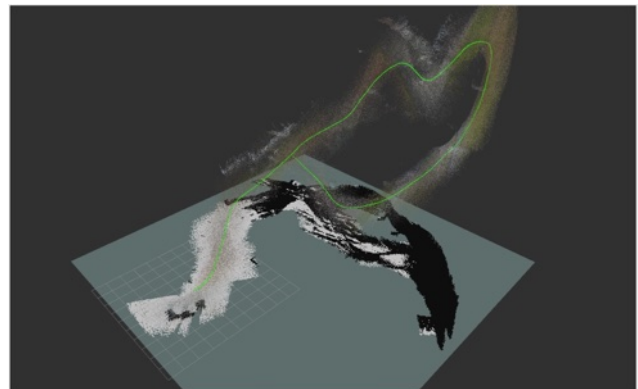


Figure 7.15: Map and pointcloud visualised in Rviz

based on RGB-D mapping in Courtyard

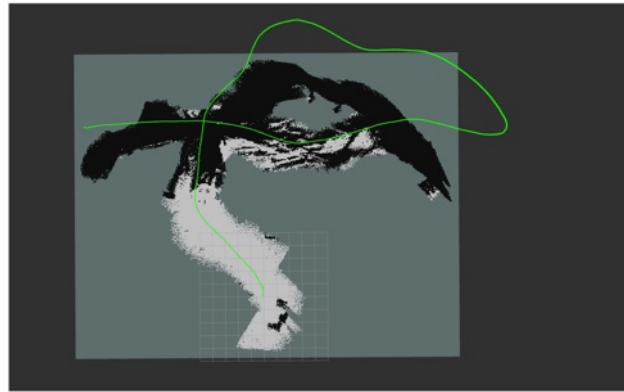


Figure 7.16: Map and path visualised in Rviz based on RGB-D mapping in Courtyard

For the courtyard tests with RGB-D odometry, as with the laboratory tests, it loses track of the position immediately, which is reflected in the remainder of the test, as seen in Figure 7.12 and 7.16. As for vertical movement, it "performs significantly worse due to the length of the test. However, it provides inconsistent results in relation to vertical movement, which is attributed to odometry failure as also illustrated in Figure 7.15 and 7.13. With the addition of GPS for ground truth, the correct path can be seen in Figure 7.12, the reliability and precision varies quite significantly in some areas, which are often close to buildings. The lowered reliability of the GPS is also seen in Figure 7.14, as this movement was not observed during the test

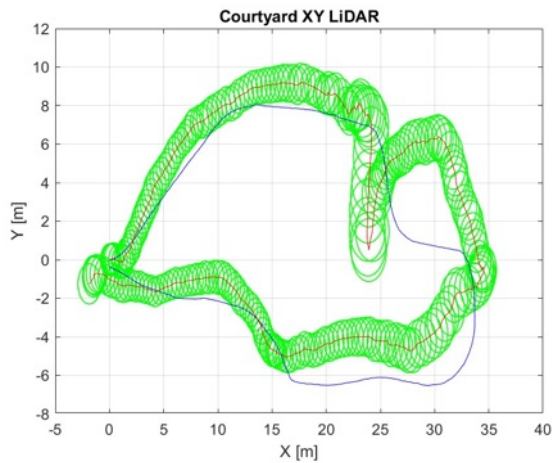


Figure 7.17: Path of the robot based on LiDAR mapping in Courtyard.

Blue line indicates RTAB-Map odometry, Red GPS mapping in Courtyard signal and green circles show 2D accuracy of the GPS

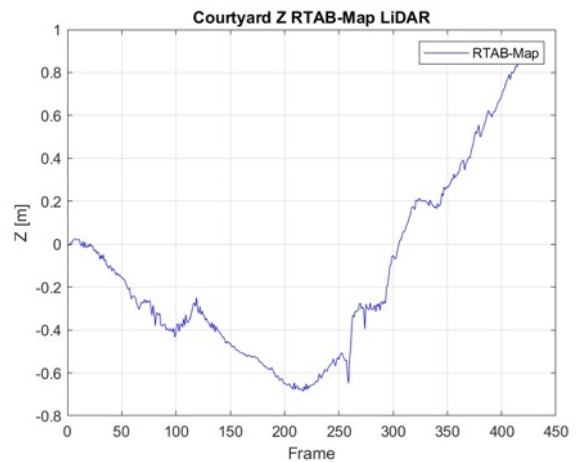


Figure 7.18: Altitude of the robot based on LiDAR

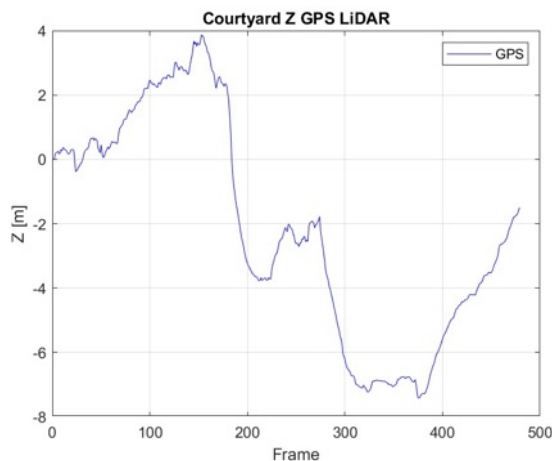


Figure 7.19: Altitude of the robot based on GPS during LiDAR mapping in Courtyard

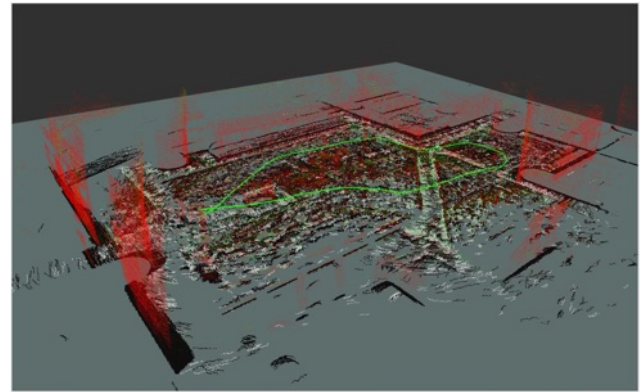


Figure 7.20: Map and pointcloud visualised in Rviz based on LiDAR mapping in Courtyard

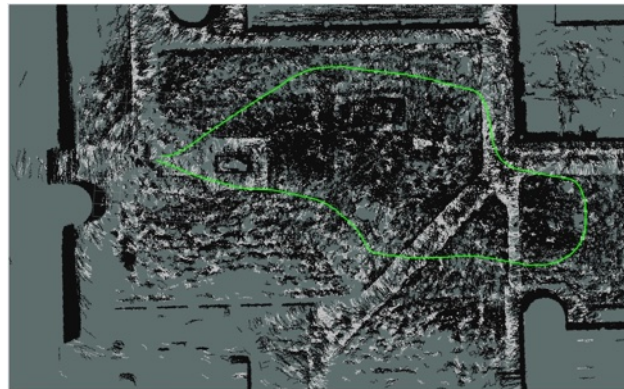


Figure 7.21: Map and path visualised in Rviz based on LiDAR mapping in Courtyard

When utilising the ICP odometry, a better performance is observed in both the XY-plane and in vertical movement as seen in Figure 7.17, 7.18 and 7.21. The LiDAR successfully detects surroundings and the results showcase that it is very efficient to utilise ICP odometry in this kind of environment. However, the GPS signal was disrupted during the test as can be seen in Figure 7.17, as this was not the correct path as showcased in both Figure 7.21 and 7.20. Therefore, the GPS was not considered viable for ground truth in this scenario

Football pitch

The football pitch, located near the Aalborg University library, was considered a significantly larger, more unstructured, and feature-sparse environment compared to the previous two locations. The tests were conducted by driving the RobuROC from one end of the pitch to the other, navigating around a small hill, and then returning to the starting position.

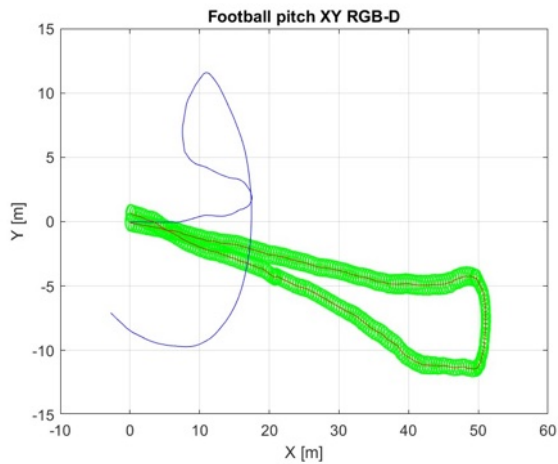


Figure 7.22: Path of the robot based on RGB-D mapping in Football pitch

Blue line indicates RTAB-Map odometry, Red GPS signal and green circles show 2D accuracy of the GPS

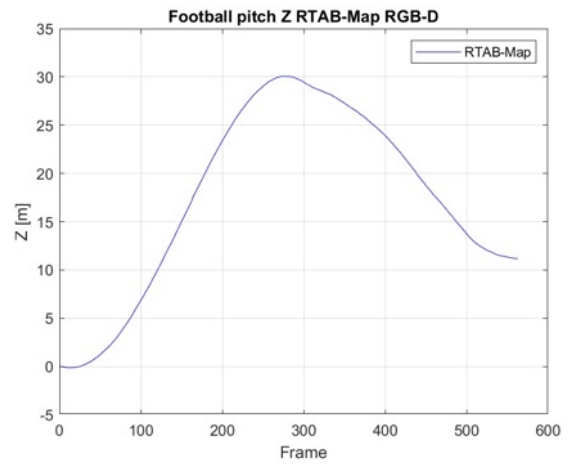


Figure 7.23: Altitude of the robot based on RGB-D

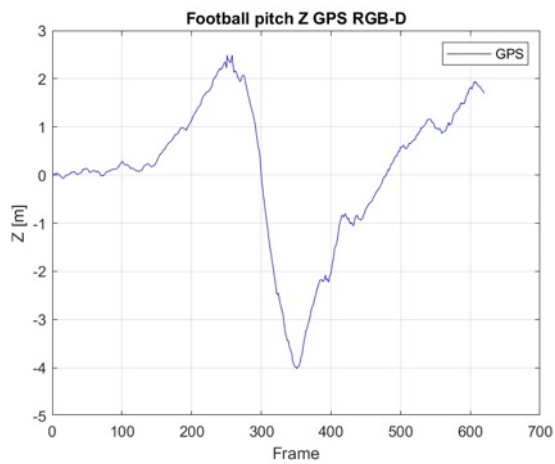


Figure 7.24: Altitude of the robot based on GPS during RGB-D mapping in Football pitch

Blue line indicates RTAB-Map odometry, Red GPS signal and green circles show 2D accuracy of the GPS

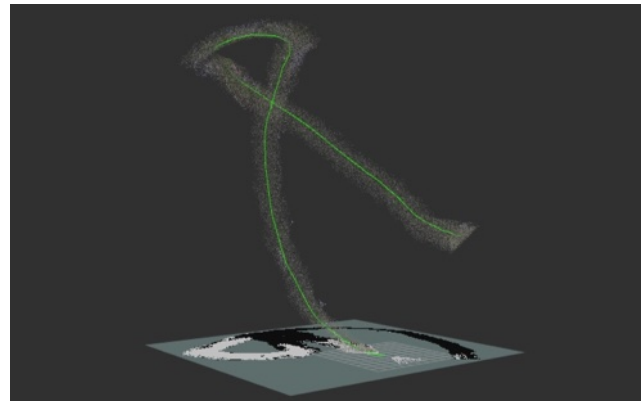


Figure 7.25: Map and pointcloud visualised in Rviz based on RGB-D mapping in Football pitch

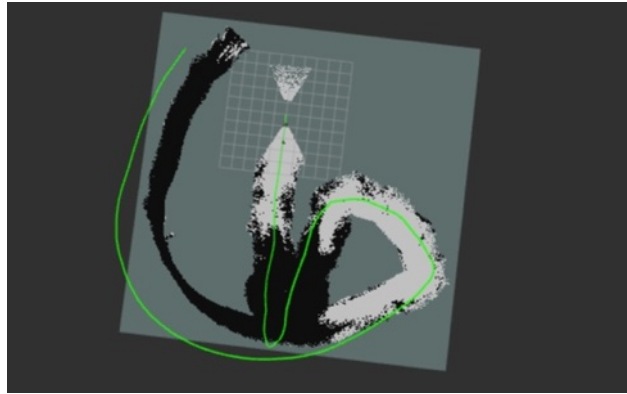


Figure 7.26: Map and path visualised in Rviz based on RGB-D mapping in Football pitch

As can be seen in Figure 7.22 and Figure 7.23, the RGB-D Odometry fails immediately as it starts out by going in the wrong direction. Similarly, upward movement is observed in the point cloud map depicted in Figure 7.25 and the data Figure 7.23, emphasizing that the RGB-D odometry encounters significant challenges in accurately handling vertical movement. Furthermore, it consistently struggles to accurately track the rate of change during turns.

Furthermore, as with the courtyard tests, the GPS does not attain satisfactory results in the Z-direction, as seen in Figure 7.24 and can therefore only be relied upon as ground truth for the XY-plane. As for the mapping module, as with the other tests utilising RGB-D odometry, fails at creating a usable map, due to the large amount of false positives in occupied space as seen in Figure 7.26.

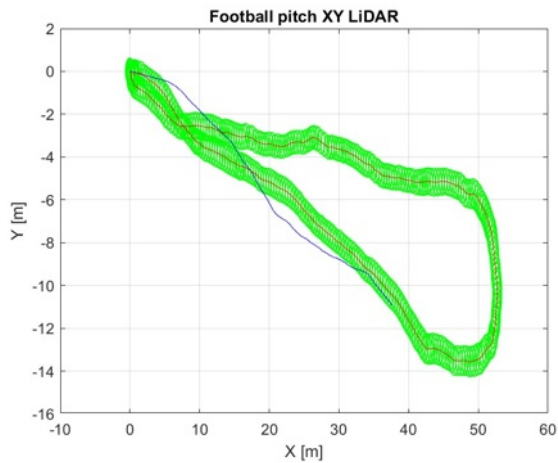


Figure 7.27: Path of the robot based on LiDAR mapping in Football pitch

Blue line indicates RTAB-Map odometry, Red GPS mapping in Football pitch signal and green circles show 2D accuracy of the GPS

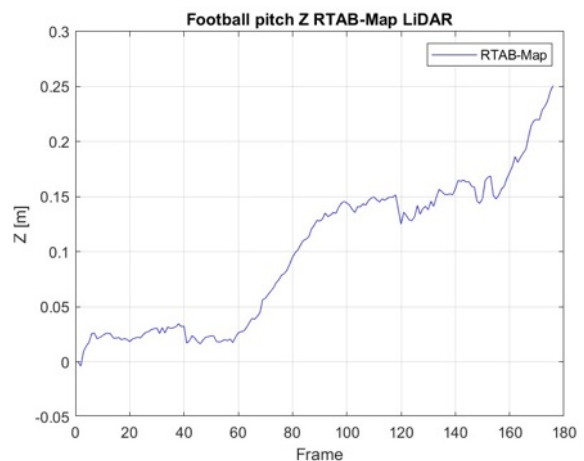


Figure 7.28: Altitude of the robot based on LiDAR

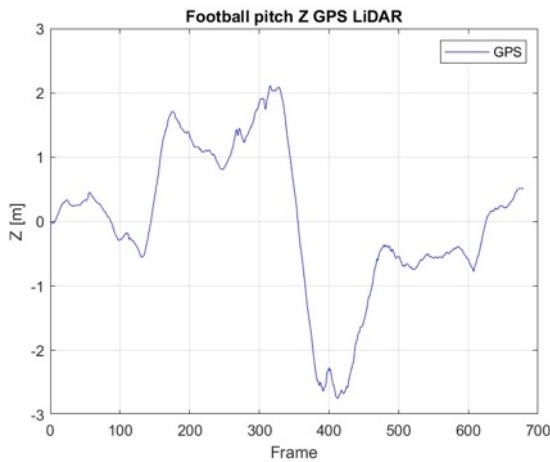


Figure 7.29: Altitude of the robot based on GPS during LiDAR mapping in Football pitch

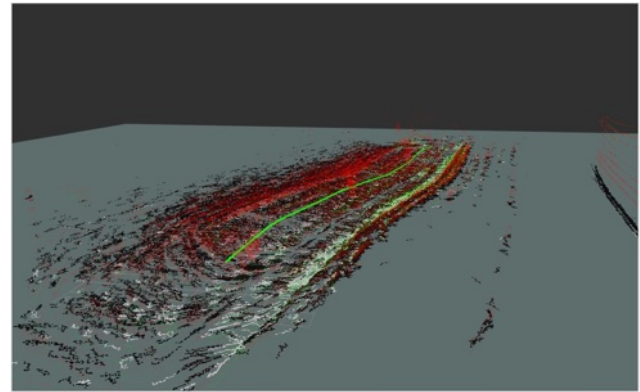


Figure 7.30: Map and pointcloud visualised in Rviz based on LiDAR mapping in Football pitch

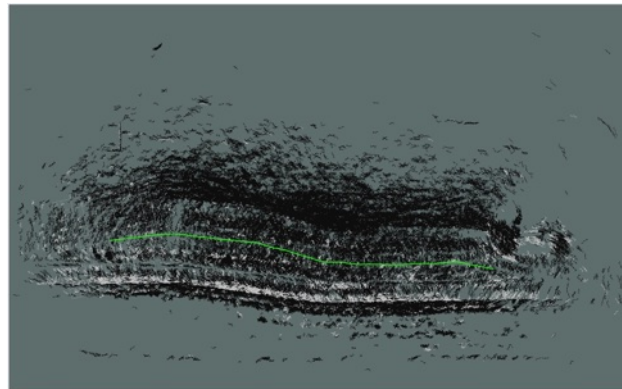


Figure 7.31: Map and path visualised in Rviz based on LiDAR mapping in Football pitch

For the ICP odometry in this case, it failed during the initial part of the test, which can be seen in Figure 7.27, 7.31 and 7.30, this is contributed to the LiDAR not providing enough data to correctly estimate odometry. However, for the short period it was functional, it provided a well defined path through a relatively correct map. Concurrently, it maintained a well defined and consistent high estimate as seen in Figure 7.28. The GPS data here is again more consistent in the XY-plane Figure 7.27 and showing less accurate data in the z-plane as seen in Figure 7.29

7.2 Tool

The purpose of this test is to verify the effectiveness of the tool and it's weight. The weight test was conducted in laboratory. The tool was dismantled and scale was used to evaluate tool's weight The effectiveness test was conducted on building site Figure 7.32.



Figure 7.32: Testing ground for the tool

The tool was mounted on the rover and lowered to ground. Then by using joystick the rover was moved around to simulate ground work. Afterwards blade and fixtures were inspected to find any deformations in metal parts, cracks in 3D printed parts or any other failures. Two variants of the tool were decided to be used for testing 90° Figure 7.33 and 50° . First configuration was tested for raking and pushing.



Figure 7.33: 90° setup of the tool

During test preparation it was noticed that the tool in normal configuration could not reach the ground. Additionally, as same problem of the tool not staying in place would occur. Therefore it was decided to flip plate 180° Figure 7.34 and use it for pushing as this setup could properly dig into the ground and be more effective in pushing.



Figure 7.34: 50° setup of the tool

7.2.1 Results

Weight

The weight of the tool was tested once after the effectiveness test of the tool. The scale showed 1846g as seen in Figure 7.35.

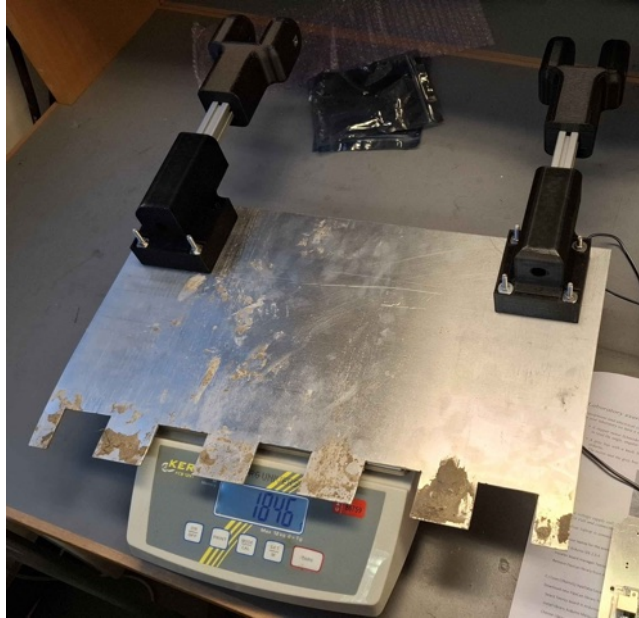


Figure 7.35: Tool weight testing setup

90° set-up

Although the tool did not brake the friction in the joint did not stop tool from moving. Therefore, testing was very limited, as tool after meeting resistance would move upward. The results of the rake can be seen in Figure 7.36.



Figure 7.36: Raking pattern created by the tool

During pushing similar event occur the plate moved when it meet any resistance until it touched the rover's front. Although plate was not in optimal position it small mound was created, which can be seen in Figure



Figure 7.37: Small mound of ground created by the tool

50° set-up

Although the tool was able to dig into the ground the forces were too great and rover could not move forward.



Figure 7.38: Attempt at pushing ground in 50° setup

After couple attempts to move forward loud crack was heard and test was stopped. After the rover was brought back to robotic laboratory the tool was disassembled and inspected. Inspection showed that one of the frame fixtures broke Figure 7.39.



Figure 7.39: Broken part of the tool fixture

8 Discussion

The development and implementation of the RobuROC 4 interface and SLAM have provided invaluable opportunity to explore challenges and considerations in Lunar explorative and formative contexts. Albeit not complete, the system also provides valuable insights into the challenges of implementing SLAM and the scene, structure and stability of the environment and their influence on the reliability and accuracy of SLAM algorithms. While the system is not complete, it does provide some of the desired functionality.

This section critically examines the system's performance, drawing on experimental results and the expectations. Key areas of analysis include the challenges and issues related to testing and platform limitations. By reflecting on these findings, this discussion seeks to identify areas for improvement for the future development and additions.

8.1 Examining the test results

This section provides an interpretation of the results obtained from the tests conducted. While the project itself does not introduce novel contributions or offer a comprehensive solution to the problem addressed, it highlights the challenges associated with implementing 3D-compatible SLAM systems. Specifically, it emphasizes the complexities of adapting algorithms to address particular issues, which often involve nuanced decisions regarding parameter selection and tuning. The current implementation of the interface and SLAM algorithm reflects these challenges.

Moreover, testing of the tool, although not shaped as desired, revealed potential areas for improvement in the design. These insights highlight the iterative nature of refining such systems and the need for further optimization to meet the intended objectives.

8.1.1 System testing

In relation to the data collected during system testing a significant amount of undesired behaviour was observed, which this section will attempt to clarify. During testing of the system, which resulted in both a relative position, a map, a point cloud of said map and a path from RTAB-Map, the individual parts of the output were analysed for the origin of the observed behaviour. Furthermore, the individual methods are evaluated as well.

Origin of undesired behaviour

Albeit not a conclusive answer was found, however there are a number of areas in which these issues could originate:

- Sensor instability and calibration
- Parameter tuning
- Platform stability

The sensors were mounted using 3D-printed parts and Velcro, which offered limited stability, particularly on irregular terrains. This instability often caused the sensors to vibrate, leading to erroneous or invalid data that interfered with RTAB-Map's odometry calculations. Additionally, while the RGB-D cameras were calibrated using their built-in on-chip calibration and the LiDAR had no calibration, more precise calibration methods could potentially mitigate some of the observed issues.

Specific challenges with RGB-D mapping and localization are likely tied to inadequacies in the feature extraction and loop closure methods, as well as suboptimal parameter tuning. For instance, the multi-RGB-D configuration did not function as expected, necessitating further exploration of the parameter space to identify and address the underlying causes. In relation to RTAB-Map, several critical limitations were identified. First, the odometry performance was particularly susceptible to sensor vibrations, which disrupted the system's ability to maintain consistent localization. Second, the RGB-D-based maps exhibited significant errors, including exaggerated vertical movements, suggesting that the default feature extraction methods and loop closure algorithms were not well-suited to the specific challenges posed by the test environments. Lastly, parameter tuning remains a significant area for improvement, as improper configurations could compromise RTAB-Map's ability to accurately interpret the data, especially when working with multi-sensor setups or in feature-sparse environments.

8.1.2 Parameter Tuning

While it was initially assumed that RTAB-Map would function effectively with its default parameters, the system failed to produce usable results. Parameter tuning and adapting frameworks and algorithms to specific tasks are crucial, yet RTAB-Map includes over 800 configurable parameters, encompassing both conditional and functionality settings. Due to the scope of this project, testing was limited to the default parameters. However, RTAB-Map offers a variety of feature extraction and graph optimization methods that warrant further investigation. These capabilities could be particularly beneficial for improving RGB-D odometry and mapping, as several parameters directly influence critical factors such as loop closure thresholds and overall system performance. Further experimentation with these configurations could yield more robust results tailored to specific scenarios.

RGB-D vs. LiDAR

In evaluation of the two methods utilised in SLAM, the results suggest that LiDAR odometry perform significantly better than RGB-D odometry. RGB-D-based odometry and mapping did not perform as intended in the control laboratory's dense and structured environment, in which it is expected to perform better, suggests that improvements are needed, as discussed above. The exaggerated vertical movements detected in RGB-D odometry influences the mapping and localisation significantly, as it overlaps with the existing map. Conversely, the LiDAR odometry in both feature-rich and dense environments, shows promise as it continuously maintain a good localisation and somewhat good mapping. However, for the LiDAR, it is believed that its range helped considerably, as it was able to consistently obtain good features from buildings.

Testing sites

Despite the fact that the testing sites were selected with the objective of assessing the limits of the system, issues were encountered in each of them that had the potential to compromise the overall results.

- **Control lab:** This environment proved to be highly dynamic as students and personnel passed through at all hours. This proved difficult to handle for LiDAR mapping which had long range and could see whole laboratory. Therefore multiple outliers were created in the pointcloud map due to people walking around the rover. That was not the case for RGB-D which not only has 3m range but was also placed at 16° angle pointing towards the ground. Furthermore, poor lighting conditions could hamper the RGB-D as reflective points on the floor sometimes was seen in the Rviz as holes.

- **Courtyard:** Although courtyard was bigger than laboratory for LiDAR it had no difference due to its range. Additionally, lighting were better, but reliant on the weather. Ideally this environment should be replaced with an open field with some obstacles in the middle as LiDAR could easily take surrounding buildings as reference for map building and localising.
- **Football pitch:** The football pitch was chosen to be similar environment to what the actual lunar rover would encounter. The environment was considered satisfactory, due to the proximity of the area, here considered battery lifetime for transport and general resources made available. The row of trees along pavement did contributed to LiDAR effectiveness and driving away from them made probably it lost.

8.2 Sensor Placement

The sensor placement was chosen based on having as large a field of view as possible and providing full coverage of objects and obstacles when traversing them or moving around them. However, this might not have been the best option for the SLAM method utilised. These considerations are discussed below.

RGB-D cameras

In relation to the placement of the RGB-D realsense D435i cameras, was considered fine until testing, where it became apparent that RTAB-Map considers forward movement to be slightly upwards. This, as discussed earlier, influences its reliability and usability for this project and therefore, the placement of the cameras should either be reconsidered or SLAM should be adapted to accommodate for this placement. Since the cameras are pointed towards the ground, while driving, the flow of the images is upwards, which potentially interferes with RTAB-Map odometry. As some features had the correct flow the output vector of movement could be pointing at an angle forward. Although the angle of the cameras was included in the frame of the robot description, RTAB-Map might not have included this information properly or the resulting angle from RGB-D odometry was higher than the angle of the cameras.

Lidar

The placement of LiDAR was satisfactory as the FoV of the sensor was not obstructed by the rover and the ground coverage near the rover was also good. The drawback to that solution was the inability to place the provided cover on the robot to protect exposed electronics from the weather. If the LiDAR was placed on top of the cover, the detectable area around the robot would be insufficient.

Tool testing

The tool in 90° configuration was proved to some extend effective. However, further development is required to optimise its performance. The primary issue identified was the instability of the hinge mechanisms, which was found to be a significant contributing factor to the tool's overall performance. Moreover, the tool measurements were taken on flat solid ground. Due to that the arms of the tool were slightly too short and due to uneven terrain only parts of the tool were engaged by the ground. In order to improve the tool some sort of locking mechanism should be introduced and aluminium arms should be extended by approximately 10cm.

8.3 Other considerations

The biggest issue during testing was memory issues on the NUC. The ROS bags were taking too much space resulting in only one or two tests could be conducted before all the space on the drive was occupied. The attempt to solve this issue was to record data directly on an external hard drive. Although it solved the issue it also made the rover less responsive as the communication between rover and gamepad was disrupted by the data transfer. Therefore, the only solution was to save the data on the external drive after test was conducted.

Another aspect of the system was the effectiveness of RTAB-Map. It was chosen due to the available good documentation and ability to use multiple cameras. Although documentation helps to solve some issues that occur during implementation. The attempt to use both cameras for visual odometry failed. Using IMU or wheel odometry could help stabilising rover in space for visual odometry and help localisation with LiDAR within sparse environments. Moreover, adding IMU has two benefits; it can enhance the RGBD odometry by adding gravity constraints in the local bundle adjustment used by RTAB-Map. It can also be leveraged in the SLAM node to add gravity constraints in the graph. The whole map will then be aligned with gravity. Although SLAM node is subscribing to the RGB-D topic during lidar SLAM it seems that it does not use it. Therefore it would be advantageous to explore other SLAM algorithms such as LVI-SAM.

9 Conclusion

In conclusion, the provided platform was adapted to the needs of the project by adding additional sensors, including RGB-D cameras and a Lidar. Subsequently, an interface was implemented for easy integration with ROS2. Furthermore, a SLAM algorithm, specifically RTAB-Map, was integrated through ROS2, with default parameters and tested extensively with multiple sensor modalities. Thus, the objectives specified in chapter 3 are evaluated based on the results obtained and the progress made:

1. **Objective 1:** The ROS2 was integrated into the robot system through an interface. The system can receive signals from all installed sensors; two RGB-D cameras, a LiDAR and an IMU. Additionally, the CANOpen protocol was used within the interface communicate with the actuators and a DUALSHOCK controller. From the perspective of the group, this objective is considered fulfilled.
2. **Objective 2:** The RTAB-Map was implemented and was used to map three different environments with decreasing feature density. In rich and densely featured environments, laboratory and courtyard environments respectively, the LiDAR-based odometry performed well. The Lidar odometry performed well, while RGB-D odometry had issues with keeping track. It was shown that less features in the environment makes mapping and localising challenging and in feature-sparse conditions the SLAM algorithm failed to localise correctly, which resulted in the system crashing. From the perspective of the group, this objective is considered partially fulfilled.
3. **Objective 3:** No navigation and path planning algorithms has been introduced to the system. Therefore, from the perspective of the group, this objective is not considered fulfilled.
4. **Objective 4:** A tool was designed, constructed and mounted in the rover. The tool was subsequently tested and evaluated based on its performance. Based on the results some additions to the design are proposed. From the perspective of the group, this objective is considered partially fulfilled.
5. **Objective 5:** The system was tested and evaluated based on the gathered data. Although the environments it was tested within might not all have been optimal. Considering the delimitations of the project, this objective is, from the perspective of the project group considered fulfilled.

9.1 Future Works

The primary challenge in a scarce environment was the lack of available features which could be utilised for localisation. An implementation wherein wheel odometry or integrating an IMU, could potentially improve the results obtained. Furthermore, if another solution was chosen or RTAB-Map was solely used for mapping an integration of an Extended Kalman Filter.

Another option for improving the system would be to tune the parameters of RTAB-Map to improve or change the mode of operation, since the options for this are many. Furthermore, RTAB-Map is capable of creating 2.5D maps, and thus the obtained cloud points could be processed externally to obtain a 3D map.

Alternatively, another SLAM algorithm could be implemented, such as LVI-SAM. Additionally, path planning in 3D should be implemented in order to make the platform fully autonomous. Furthermore, the current tool should be redesigned to become more effective and more tools could be implemented, such as a compactor and a ripper.

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Appendices

A Github repository

The github repository for this project can be found at:

<https://github.com/Blueguardian/P9-RobuROC4>